Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Thesis for the Master's degree in Physics

Abstract

[To be continued...]

- Appendix on s-wave SC ghost simulations on HM
- Set ComputerModern font in all CairoMakie figures

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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PH	Particle-hole
PHS	Particle-hole symmetry
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued…]

Chapter 1

Introduction to the Extended Hubbard Model

A major breakthrough in Condensed Matter Physics was the discovery of superconductivity in two-dimensional copper-oxide based compounds (often referred to as "cuprates"). Until 1986, when Bednorz and Muller found evidence of a superconducting transition in Ba-La-Cu-O systems [5], superconductivity had been observed up to critical temperatures (T_c) around 30 K. The new class of copper oxides rapidly took over as highest-T_c class of superconductors, dramatically increasing critical temperatures to around 130 K at ambient pressure. Today, the highest observed value is $T_c \approx 151$ K at ambient pressure on the cuprate $HgBa_2Ca_2Cu_3O_{8+\delta}$ [6]. It is worth mentioning that this result was obtained in recent months after almost 30 years from the previous record.

Cuprates Physics is particularly interesting and surprising also considering their very poor conducting properties at normal phase. Copper oxides are basically ceramics, highly prone to establishing insulating antiferromagnetic phases. Antiferromagnetism and superconductivity are apparently antithetical orderings: the first arises from electronic repulsion, while the second from attractive interactions [7]. It is widely observed, however, how in cuprates the two phases can compete **and even coexist** [8].

Moreover, cuprates superconduct through a new, not-yet understood mechanism. Classical Bardeen-Cooper-Shrieffer (BCS) theory of superconductivity describes the arising of a phonon-induced effective electron-electron attractive interaction as a source of so-called "conventional superconductivity". It is the interplay of electrons with lattice vibrations that couples them in Cooper pairs. The same theory fails completely in explaining this new "unconventional superconductivity". All of these observation have gathered, in years, a lot of interest in these materials.

Such complex materials are surprisingly well described in many features by the simple (but rich) single-band Hubbard model (HM) [8],

$$\hat{H}_{\mathrm{HM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \qquad t, U > 0 \tag{1.1}$$

being t the hopping energy and U the on-site repulsion. Sum is intended on the spin index $\sigma \in \{\uparrow, \downarrow\}$ and the sites i of a two-dimensional square lattice. The notation $\langle ij \rangle$ indicates nearest neighbours (NNs). Recent developments [9] have shown, however, that a more accurate description of these materials could be obtained by adding a non-local electron-electron attraction. This leads us to a widely studied modified version of Eq. (1.1),

$$\hat{H}_{\mathrm{EHM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \qquad t, U, V > 0 \tag{1.2}$$

commonly referred to as "Extended Hubbard model" (EHM). The difference with the conventional HM is the presence of a NNs attractive interaction, independent of site and spin indices. In this thesis, we study the insurgence of antiferromagnetism and superconductivity in the EHM due to the presence of this new interaction. To do so, we use numerical Hartree-Fock (HF) methods.

This introductory chapter is organised as follows: [To be continued...]

Chapter 2

Introduction to Hartree-Fock methods

In this chapter we give a general introduction to the Hartree-Fock (HF) variational method and its numerical implementation. In Quantum Mechanics, variational methods allow to approximate the ground state of a many-body hamiltonian by minimising the energy computed for a specific substate of physical states. In particular, in the HF method we approximate the real ground-state with the best possible many-body state that is an eigenstate of a non-interacting fermionic hamiltonian.

Despite its simplicity, the method is very powerful for describing various types of quantum many-body phenomena and, in particular, spontaneous symmetry breaking (SSB). The description of HF method in this chapter is largely based on [29, 31].

2.1 Variational principle and gaussian states

Let $\hat{H}$ be a generic hamiltonian acting on the Hilbert space $\mathcal{H}$. For any many-body state $|\Psi\rangle \in \mathcal{H}$, the energy is given by the expectation value (EV)

$$E[\Psi] = \langle\Psi|\hat{H}|\Psi\rangle$$

Let $|\Psi_0\rangle$ be the ground state of $\hat{H}$, and $E_0 \equiv E[\Psi_0]$. Then

$$\forall\, |\Psi\rangle \in \mathcal{H} \quad : \quad E[\Psi] \geq E_0$$

Let $\mathcal{S} \subset \mathcal{H}$ be a subspace of states, ad let $|\Psi\rangle$ be a state inside that locally minimises energy,

$$|\Psi\rangle \in \mathcal{S} \qquad \frac{\delta E[\Psi]}{\delta\, |\Psi\rangle} = 0$$

If the true ground state belongs to $\mathcal{S}$, the found local minimum coincides with $|\Psi_0\rangle$; otherwise, approximates it. In Fig. 2.1 we give a pictorial representation of this idea: the state $|\Psi\rangle \in \mathcal{S}$ is an approximation, found by constrained minimisation, of the true ground-state.

In HF methods, we identify $\mathcal{S}$ with the set of so-called "gaussian states". These are all the ground-states of quadratic hamiltonians, namely the hamiltonians that are bilinear in the fermionic creations operators (see App. **??** for details). The restriction to gaussian states significantly simplifies the computation of expectation values. Indeed, thanks to Wick's theorem, the expectation value of quartic operators exactly decomposes as

$$\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\delta\rangle\langle \hat{c}^\dagger_\beta \hat{c}_\gamma\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\beta\rangle\langle \hat{c}^\dagger_\gamma \hat{c}_\delta\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta\rangle\langle \hat{c}_\gamma \hat{c}_\delta\rangle}_{\text{Cooper}} \tag{2.1}$$

The subscripts indicate the standard names to the different types of contractions, that we adopt throughout the text.

Operatively, one can think of the HF merhod as an approximation of the the hamiltonian with a non-interacting hamiltonian

$$\hat{H} \overset{\text{HF}}{\simeq} \hat{H}_{\text{HF}} = \sum_\alpha E_\alpha \hat{\gamma}^\dagger_\alpha \hat{\gamma}_\alpha$$

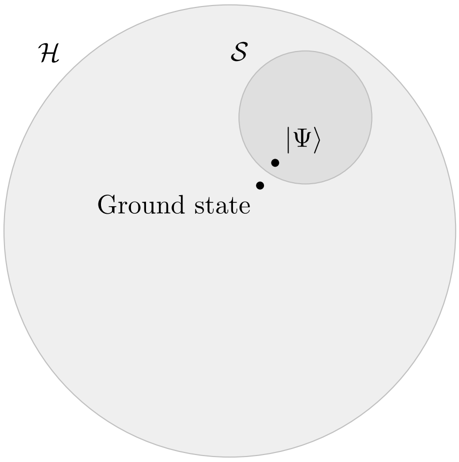


Figure 2.1 | Pictorial representation of the search for the best approximation $|\Psi\rangle$ to the true ground-state inside the subspace $\mathcal{S} \subset \mathcal{H}$.

where $\hat{\gamma}_\alpha$ are fermionic operators determined by the minimisation procedure

$$\left\{\hat{\gamma}_\alpha, \hat{\gamma}_\beta^\dagger\right\} = \delta_{\alpha\beta}$$

and are connected to the conventional electronic operators $\hat{c}_\alpha$ through an unitary map. See App. **??** for details.

Let us stress that the operatorial approximation $\overset{\mathrm{HF}}{\simeq}$ is a shorthand notation that must be intended as follows: $\hat{H}_{\mathrm{HF}}$ is that non-interacting operator whose ground-state is the best approximation to the ground-state of $\hat{H}$. This does not mean that in general the two operators are related.

2.2 The HF variational method

We now dicuss the implementation of the variational HF method for the generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha,\beta} A_{\alpha\beta} \hat{c}_\alpha^\dagger \hat{c}_\beta + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \tag{2.2}$$

where

$$A_{\alpha\beta} = A_{\beta\alpha}^* \qquad B_{\alpha\beta\gamma\delta} = B_{\delta\gamma\beta\alpha}^* \tag{2.3}$$

to ensure hermiticity. Moreover, the following equation must hold:

$$B_{\alpha\beta\gamma\delta} = B_{\beta\alpha\delta\gamma} \tag{2.4}$$

due to the property $\hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta = \hat{c}_\beta^\dagger \hat{c}_\alpha^\dagger \hat{c}_\delta \hat{c}_\gamma$, which follows from Fermi anticommutation rules.

Our aim is to approximate the above hamiltonian with a non-interacting hamiltonian. The most general quadratic hamiltonian is given by:

$$\hat{H} \simeq \hat{H}_{\mathrm{HF}} \equiv E_0 + \sum_{\alpha,\beta} k_{\alpha\beta} \hat{\gamma}_\alpha^\dagger \hat{\gamma}_\beta \tag{2.5}$$

As is specified in Sec. **??**, γ and c operators must be linked by an unitary map. To find the best approximation to the original hamiltonian means to identify:

- the offset E_0;
- the fermionic operators $\hat{\gamma}_\alpha$, $\hat{\gamma}_\alpha^\dagger$;
- the $k_{\alpha\beta}$ coefficients.

Note that, in terms of algebraic solution, E_0 is irrelevant. To determine it correctly is only relevant in order to get the correct approximation of the actual ground-state energy. Let us discuss briefly the features of $\hat{H}_{\mathrm{HF}}$ and then propose a solution.

2.2.1 Properties of the target quadratic hamiltonian

In this section we enumerate some useful properties of $\hat{H}_{\text{HF}}$. We will use also the following notation:

$$\hat{a}^-_\alpha \equiv \hat{c}_\alpha \qquad \hat{a}^+_\alpha \equiv \hat{c}^\dagger_\alpha$$

Due to linearity of the unitary mapping, the operator $\hat{H}_{\text{HF}}$ of Eq. (2.5) can also be expressed as the most general quadratic hamiltonians for electronic operators:

$$\hat{H}_{\text{HF}} = E_0 + \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \hat{a}^p_\alpha \hat{a}^q_\beta \tag{2.6}$$

By changing basis from $\hat{\gamma}$ to $\hat{c}$ operators, we moved all our ignorance into the new $h^{pq}_{\alpha\beta}$ matrix elements. These satisfy the relations:

$$h^{++}_{\alpha\beta} = \left(h^{--}_{\beta\alpha}\right)^* \tag{2.7}$$

$$h^{+-}_{\beta\alpha} = -h^{-+}_{\alpha\beta} \tag{2.8}$$

Because of these properties, we actually need to determine just half of the fields, since the other half will be uniquely determined.

Proof 2.1 (Symmetry properties of the tensor $h^{pq}_{\alpha\beta}$).

Let us start by proving Eq. (2.7). The HF hamiltonian can be represented through Nambu spinors as

$$\begin{aligned}\hat{H}_{\text{HF}} &= E_0 + \hat{\boldsymbol{\Phi}}^\dagger \mathbf{h} \hat{\boldsymbol{\Phi}} \\ &= E_0 + \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}^\dagger \begin{bmatrix} h^{+-} & h^{++} \\ h^{--} & h^{-+} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}\end{aligned}$$

Our hamiltonian must be hermitian. We conclude that:

$$\hat{H}^\dagger_{\text{HF}} = \hat{H}_{\text{HF}} \qquad \Longrightarrow \qquad h^{++} = \left[h^{--}\right]^\dagger$$

which reads, element-wise:

$$h^{++}_{\alpha\beta} = \left(h^{--}_{\beta\alpha}\right)^*$$

This proves Eq. (2.7).

Eq. (2.8) is proven as follows. We can set, without loss of generality,

$$\operatorname{Tr} h^{+-} + \operatorname{Tr} h^{-+} = 0$$

This passage is guaranteed by the fact that if $\mathbf{h}$ is not traceless, we can redefine it by subtracting its trace and including the latter in the energy shift E_0. Then

$$\begin{aligned}\hat{H}_{\text{HF}} &= E_0 + \sum_{\alpha,\beta} \left[h^{+-}_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + h^{-+}_{\alpha\beta} \hat{c}_\alpha \hat{c}^\dagger_\beta\right] + (\text{same sign terms}) \\ &= E_0 + \sum_{\alpha,\beta} \left[h^{+-}_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{c}_\beta \hat{c}^\dagger_\alpha\right) + h^{-+}_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{c}^\dagger_\beta \hat{c}_\alpha\right)\right] + (\text{same sign terms}) \\ &= E_0 - \sum_{\alpha,\beta} \left[h^{+-}_{\beta\alpha} \hat{c}_\alpha \hat{c}^\dagger_\beta + h^{-+}_{\beta\alpha} \hat{c}^\dagger_\alpha \hat{c}_\beta\right] + (\text{same sign terms})\end{aligned}$$

In second row we got rid of the traces sum and exchanged indices $\alpha \leftrightarrow \beta$. Confronting last and first row we conclude:

$$h^{+-}_{\beta\alpha} = -h^{-+}_{\alpha\beta}$$

which proves Eq. (2.8).

These properties were all we needed to get to the main argument of the method.

2.2.2 Self-consistent variational solution

This section gives the operational method to obtain the variational solution. We follow and extend the argument proposed by Fabrizio [31]. Let:

$$\Delta^{pq}_{\alpha\beta} \equiv \left\langle \hat{a}^p_\alpha \hat{a}^q_\beta \right\rangle$$

thus, explicitly

$$\Delta^{+-}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha \hat{c}_\beta \right\rangle \qquad \Delta^{-+}_{\alpha\beta} = \left\langle \hat{c}_\alpha \hat{c}^\dagger_\beta \right\rangle \qquad \Delta^{++}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \right\rangle \qquad \Delta^{--}_{\alpha\beta} = \left\langle \hat{c}_\alpha \hat{c}_\beta \right\rangle \tag{2.9}$$

The first two are "normal densities"; the second two are "anomalous densities". The reason for such names is the fact that these expectation values are non-zero if total charge conservation is broken. Conjugation of the last two gives a symmetry relation similar to Eq. (2.7):

$$\left(\Delta^{++}_{\alpha\beta}\right)^\dagger = \Delta^{--}_{\beta\alpha} \tag{2.10}$$

Here $\Delta^{pq}_{\alpha\beta}$ are functions of the fields $h^{pq}_{\alpha\beta}$. These quantities satisfy the equation:

$$\sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta} \frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = 0 \tag{2.11}$$

Let us prove this.

Proof 2.2 (Proof of Eq. (2.11)).

The HF ground-state energy is given by:

$$\begin{aligned} E_{\mathrm{HF}} &= \left\langle \hat{H}_{\mathrm{HF}} \right\rangle \\ &= E_0 + \sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\Delta^{pq}_{\alpha\beta} \end{aligned}$$

where the expectation value is taken over the (up to now unknown) HF ground-state. Derive it:

$$\begin{aligned} \frac{\partial E_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} &= \frac{\partial}{\partial h^{rs}_{\mu\nu}} \sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\Delta^{pq}_{\alpha\beta} \\ &= \Delta^{rs}_{\mu\nu} + \sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta} \frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \end{aligned} \tag{2.12}$$

Note that by construction E_{HF} is greater than the actual ground-state energy, since our approximated wavefunction is not the actual ground-state wavefunction. From Hellman-Feynman theorem [29], we know that for a parametric hamiltonian it holds:

$$\begin{aligned} \frac{\partial E_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} &= \langle \Psi | \frac{\partial \hat{H}_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} | \Psi \rangle \\ &= \langle \Psi | \frac{\partial}{\partial h^{rs}_{\mu\nu}} \sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta} \hat{a}^p_\alpha \hat{a}^q_\beta | \Psi \rangle \\ &= \left\langle \hat{a}^r_\mu \hat{a}^s_\nu \right\rangle \\ &= \Delta^{rs}_{\mu\nu} \end{aligned} \tag{2.13}$$

Eqns.(2.12) and (2.13), combined, imply:

$$\sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta} \frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = 0$$

which ends the proof.

We now use Eq. (2.11) to minimise the variational energy $E = \langle \hat{H} \rangle$. Performing Wick's contractions, the variational energy becomes

$$
\begin{aligned}
E &= \langle \hat{H} \rangle \\
&= \sum_{\alpha,\beta} A_{\alpha\beta} \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle \\
&= \sum_{\alpha,\beta} A_{\alpha\beta} \langle \hat{a}^+_\alpha \hat{a}^-_\beta \rangle + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \left(\langle \hat{a}^+_\alpha \hat{a}^+_\beta \rangle \langle \hat{a}^-_\gamma \hat{a}^-_\delta \rangle - \langle \hat{a}^+_\alpha \hat{a}^-_\gamma \rangle \langle \hat{a}^+_\beta \hat{a}^-_\delta \rangle + \langle \hat{a}^+_\alpha \hat{a}^-_\delta \rangle \langle \hat{a}^+_\beta \hat{a}^-_\gamma \rangle \right) \\
&= \sum_{\alpha,\beta} A_{\alpha\beta} \Delta^{+-}_{\alpha\beta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \left(\Delta^{++}_{\alpha\beta} \Delta^{--}_{\gamma\delta} - \Delta^{+-}_{\alpha\gamma} \Delta^{+-}_{\beta\delta} + \Delta^{+-}_{\alpha\delta} \Delta^{+-}_{\beta\gamma} \right) \\
&= \sum_{\alpha,\beta} A_{\alpha\beta} \Delta^{+-}_{\alpha\beta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\alpha\beta} \Delta^{+-}_{\gamma\delta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \Delta^{++}_{\alpha\beta} \Delta^{--}_{\gamma\delta}
\end{aligned}
\tag{2.14}
$$

where in the second line, we inserted the relations (2.9). Our argument is completely contained in the following statement: the HF approximation to the real ground state it obtained as the ground state of a quadratic hamiltonian whose fields $h^{pq}_{\alpha\beta}$ satisfy the following relations:

$$
h^{+-}_{\alpha\beta} = \frac{1}{2} \left[A_{\alpha\beta} + \sum_{\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\gamma\delta} \right] \tag{2.15}
$$

$$
h^{++}_{\alpha\beta} = \sum_{\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta^{--}_{\gamma\delta}}{2} \tag{2.16}
$$

These equations, operationally, give the instructions to build the best possible fermionic non-interacting hamiltonian to reproduce our original hamiltonian. These equations are proven as follows:

Proof 2.3 (Proof of Eqns. (2.15), (2.16)).

Consider E as expressed in Eq. (2.14). Upon taking the derivative, we obtain

$$
\begin{aligned}
\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} A_{\alpha\beta} \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} &+ \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \left(\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \Delta^{+-}_{\gamma\delta} + \Delta^{+-}_{\alpha\beta} \frac{\partial \Delta^{+-}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} \right) \\
&+ \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \left(\frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \Delta^{--}_{\gamma\delta} + \Delta^{++}_{\alpha\beta} \frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} \right)
\end{aligned}
$$

which, by using Eq. (2.4), becomes:

$$
\begin{aligned}
\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} &\left[A_{\alpha\beta} + \sum_{\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\gamma\delta} \right] \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \\
&+ \sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta^{--}_{\gamma\delta}}{2} \frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + \sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta^{++}_{\alpha\beta}}{2} \frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}
\end{aligned}
$$

Let us define

$$
C_{\alpha\beta} \equiv \frac{1}{2} \left[A_{\alpha\beta} + \sum_{\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\gamma\delta} \right] \qquad D_{\alpha\beta} \equiv \sum_{\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta^{--}_{\gamma\delta}}{2} \tag{2.17}
$$

From Fermi anticommutation rules, we have

$$\Delta^{+-}_{\alpha\beta} = \delta_{\alpha\beta} - \Delta^{-+}_{\beta\alpha} \qquad \Longrightarrow \qquad \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = -\frac{\partial \Delta^{-+}_{\beta\alpha}}{\partial h^{rs}_{\mu\nu}}$$

which implies

$$\begin{aligned}\sum_{\alpha,\beta} 2C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} &= \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right)\\ &= \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha}\frac{\partial \Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right)\end{aligned}$$

Moreover, using Eqns. (2.3), (2.10),

$$\sum_{\alpha,\beta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2} = \sum_{\alpha,\beta}\left(\frac{B_{\delta\gamma\beta\alpha}\Delta^{--}_{\beta\alpha}}{2}\right)^*$$

and, recognising $D_{\gamma\delta}$ from Eq. (2.17), we conclude

$$\sum_{\alpha,\beta,\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2}\frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} = \sum_{\gamma,\delta} D^*_{\delta\gamma}\frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}$$

Recollect everything and impose the solution to be a stationary point of energy,

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} \overset{!}{=} 0$$

The notation "$\overset{!}{=}$" indicates, throughout the thesis, that we are imposing some condition. In the end we have:

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha}\frac{\partial \Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D_{\alpha\beta}\frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D^*_{\beta\alpha}\frac{\partial \Delta^{--}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right) \overset{!}{=} 0 \tag{2.18}$$

Comparing this equation with Eq. (2.11). It must hold:

$$h^{+-}_{\alpha\beta} = C_{\alpha\beta} \qquad h^{++}_{\alpha\beta} = D_{\alpha\beta}$$

Recalling their explicit form from Eq. (2.17), we prove Eqns. (2.15), (2.16).

We notice that the right-hand side of both Eqns. (2.15), (2.16) depends on the fields themselves, as $\Delta^{pq}_{\alpha\beta}$ are defined as expectation values of two-points fermionic operators over the ground-state $|\Psi\rangle$. For a given set of fields $h^{pq}_{\alpha\beta}$, the latter is determined (up to some unitary transformation) as the ground-state of a fermionic non-interacting model. In other words, being $\mathbf{h}$ the tensor containing all fields,

$$\Delta^{pq}_{\alpha\beta}[\mathbf{h}] = \langle\Psi[\mathbf{h}]|\hat{a}^p_\alpha\hat{a}^q_\beta|\Psi[\mathbf{h}]\rangle \tag{2.19}$$

The three Eqns. (2.15), (2.16) and (2.19) define the solution of the HF variational method. Specifically, we need to determine $\Delta^{pq}_{\alpha\beta}$ such that, for given fields, Eq. (2.19) is respected.

The list of parameters $\{\Delta^{pq}_{\alpha\beta}\}$ to be determined self-consistently may be not that short: by looking at Eq. (2.19), the indices α, β, p, q are spanning a large Hilbert space. However, this equation can be manipulated and transformed, reducing the set of parameters to be determined self-consistently to a very manageable number. For each phase we define these last as "Hartree-Fock parameters" (HFPs): a set of objects self-consistently determined by a set equivalent, but transformed, version of Eq. (2.19).

The great advantage of this scheme is that, with an appropriate choice of the Hilbert space basis (for example, moving to reciprocal space for a translationally-invariant model), we can implement easily model symmetries.

2.2.3 Operative hamiltonian approximation

In this section we give an alternative expression for Eqns. (2.15), (2.16), by using an equivalent (but more straightforward) operatorial approximation. For a generic two-body quartic operator, the HF approximation is given by the decomposition

$$
\begin{aligned}
\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\beta \hat{c}_\alpha \overset{\text{HF}}{\simeq} & \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \langle \hat{c}_\beta \hat{c}_\alpha \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \hat{c}_\beta \hat{c}_\alpha - \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\beta \hat{c}_\alpha \rangle && \text{(Hartree)} \\
& - \hat{c}^\dagger_\alpha \hat{c}_\beta \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \hat{c}^\dagger_\beta \hat{c}_\alpha + \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle && \text{(Fock)} \\
& + \hat{c}^\dagger_\alpha \hat{c}_\alpha \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \hat{c}^\dagger_\beta \hat{c}_\beta - \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle && \text{(Cooper)}
\end{aligned}
$$

The proof of this statement is given in App. **??**, where we also argue the interpretation of HF method as a mean-field theory (MFT). Let us summarise this section through a practical rule to decompose the quartic part of the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$ in the EHM, by applying Eq. (**??**).

Local repulsion. Applying the HF method, the local repulsion $\hat{H}_U$ is approximated by the quadratic operator:

$$
\begin{aligned}
\hat{H}_U &= U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} && (2.20) \\
&\overset{\text{HF}}{\simeq} U \sum_i \Big(\hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} \rangle + \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} \rangle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} - \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} \rangle \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} \rangle && \text{(Hartree)} \\
&\qquad - \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\downarrow} \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\uparrow} \rangle - \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\downarrow} \rangle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\uparrow} + \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\downarrow} \rangle \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\uparrow} \rangle && \text{(Fock)} \\
&\qquad + \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \langle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} \rangle + \langle \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \rangle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} - \langle \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \rangle \langle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} \rangle \Big) && \text{(Cooper)}
\end{aligned}
$$

Non-local attraction. Applying the HF method, the local repulsion $\hat{H}_V$ is approximated by the quadratic operator:

$$
\begin{aligned}
\hat{H}_V &= -V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} && (2.21) \\
&\overset{\text{HF}}{\simeq} -V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \Big(\hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} \rangle + \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \rangle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} \rangle && \text{(Hartree)} \\
&\qquad - \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \rangle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} + \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \rangle \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} \rangle && \text{(Fock)} \\
&\qquad + \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \langle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} \rangle + \langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \rangle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} - \langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \rangle \langle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} \rangle \Big) && \text{(Cooper)}
\end{aligned}
$$

We will use these decomposition referring to each sector in dedicated chapters.

2.3 The iHF algorithm

In this section we describe the algorithm we used to compute Eq. (2.19). We refer to this technique as "iterative Hartree Fock" (iHF). Let $\mathbf{v}$ be the vector containing all HFPs. Suppose we have N HFPs v_i

$$
\mathbf{v} \in \mathbb{C}^N \qquad \mathbf{v} \equiv \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}
$$

In order to get a physical solution, each HFP must satisfy a self-consistency equation similar to Eq. (2.19). Each equation has the structure:

$$
v_i = (\text{Some non-linear function of } \mathbf{v})
$$

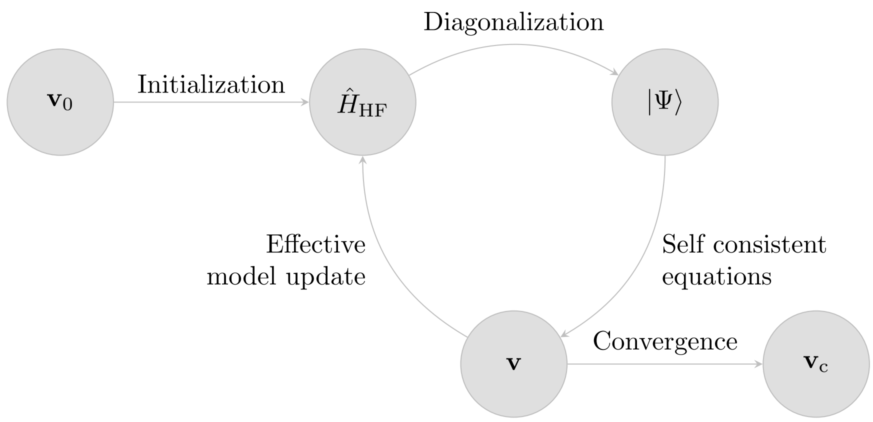


Figure 2.2 | Schematisation of the self-consistent Hartree Fock method described in Sec. 2.3. The effective model is solved, the solution is used to estimate self-consistently the Hartree Fock vector $\mathbf{v}$, which is then used to update iteratively the model itself. The notation $\mathbf{v}_0$ indicates the initialiser HFPs vector, $\mathbf{v}_\mathrm{c}$ the converged HFPs vector.

Let A be the non-linear operator collecting all these functions. The problem of finding a self-consistent solution can be reformulated as follows: $\mathbf{v}$ represents a physical solution if it is a fixed point of the map A,

$$\mathrm{A}\colon \mathbb{C}^N \to \mathbb{C}^N \qquad \mathrm{A}(\mathbf{v}) = \mathbf{v} \quad \Longrightarrow \quad \mathbf{v} \text{ is a solution.} \tag{2.22}$$

We implement the search for a self-consistent solution as follows: first, we obtain an analytic expression for A. Then we choose an initialiser $\mathbf{v}_0$ and apply A to it, obtaining a new HFPs vector. Iterating this procedure, we "follow" the non-linear map A until we converge. Specifically:

1. We start with a generic quartic hamiltonian:
$$\hat{H} = \sum_{\alpha\beta} A_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta\gamma\delta}\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta$$
We choose the model parameters and set the electronic density target;

2. Following the discussion of Sec. 2.2, we approximate analytically $\hat{H}$ with a quadratic hamiltonian
$$\hat{H} \simeq \hat{H}_{\mathrm{HF}} \equiv \sum_{\alpha\beta} \left[h^{+-}_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}_\beta - h^{+-}_{\beta\alpha}\hat{c}_\alpha \hat{c}^\dagger_\beta + h^{++}_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta + \left(h^{++}_{\beta\alpha}\right)^* \hat{c}_\alpha \hat{c}_\beta \right]$$
with the fields given by Eqns. (2.15), (2.16);

3. We impose selection rules on the fields $h^{+-}_{\alpha\beta}$, $h^{++}_{\alpha\beta}$, eventually eliminating those not respecting the symmetry sector we are targeting. For example, if we choose not to break total charge conservation, set $\forall \alpha, \beta\colon h^{++}_{\alpha\beta} = 0$;

4. We choose a starting value for the HFPs vector $\mathbf{v}_0$;

5. We set $\mathbf{v} = \mathbf{v}_0$ and enter the iterative scheme of Fig. 2.2:

   a) Get a numeric approximation of the hamiltonian for the current value of $\mathbf{v}$, $\hat{H}_{\mathrm{HF}}[\mathbf{v}]$;
   b) Determine the value $\mu[\mathbf{v}]$ giving the desired density;
   c) Diagonalise the effective hamiltonian $\hat{H}_{\mathrm{HF}}[\mathbf{v}] - \mu[\mathbf{v}]\hat{N}$, and obtain the approximated ground-state $|\Psi[\mathbf{v}]\rangle$;
   d) Compute $\mathrm{A}(\mathbf{v})$ using self-consistency equations;
   e) If the algorithm converged, exit; otherwise, update the value of $\mathbf{v}$ repeat from point a).

In the following subsections we describe technical details concerning the chemical potential fixing and the convergence criteria used for the practical implementation of the above iterative scheme.

2.3.1 Fixed density simulations and the role of μ

In this thesis all derivations and simulations are carried out at fixed target density. This is done by fixing the chemical potential μ during iterations of the algorithm. This must be done iteratively: changing at each step the HFPs vector, we are actively modifying the approximate hamiltonian. Chemical potential must be determined accordingly to maintain the required density.

For a given choice of HFPs vector and chemical potential, the ground state $|\Psi[\mathbf{v},\mu]\rangle$ is uniquely determined; then we compute density as

$$n(\mathbf{v},\mu) \equiv \frac{1}{\mathcal{V}} \sum_{\alpha} \langle \Psi[\mathbf{v},\mu]| \hat{c}^{\dagger}_{\alpha} \hat{c}_{\alpha} |\Psi[\mathbf{v},\mu]\rangle$$

being $\mathcal{V}$ the volume. Suppose we fix density to a value $n_0 \in [0,1]$. At step i, naming the current HFPs vector $\mathbf{v}_i$, μ_i (the chemical potential at current step) is determined as

$$\mu_i \equiv \min_{\mu} |n(\mathbf{v}_i,\mu) - n_0| \tag{2.23}$$

The density n must be a monotonically increasing function of μ: then we can translate the above equation, which would inspire a minimisation procedure, to a much simpler node-search rule:

$$\mu_i = \mu\,|_{\delta n(\mathbf{v}_i,\mu)=0} \qquad \text{being} \qquad \delta n(\mathbf{v}_i,\mu) \equiv n(\mathbf{v}_i,\mu) - n_0$$

In order to where the function changes sign (which is, the position of μ_i) we use a basic bisection algorithm:

1. Set an arbitrary tolerance $\Delta n > 0$ and a shifting parameter $\Delta\mu > 0$;
2. Choose starting lower and upper bounds $\mu_{\mathrm{L}} < \mu_{\mathrm{U}}$;
3. Compute $\delta n(\mathbf{v}_i,\mu_{\mathrm{L}})$. Evaluate:
   - if $|\delta n(\mathbf{v}_i,\mu_{\mathrm{L}})| \le \Delta n$, set $\mu_i = \mu_{\mathrm{L}}$ and halt;
   - if $\delta n(\mathbf{v}_i,\mu_{\mathrm{L}}) < -\Delta n$ do nothing;
   - if $\delta n(\mathbf{v}_i,\mu_{\mathrm{L}}) > \Delta n$, shift the lower bound down, $\mu_{\mathrm{L}} \to \mu_{\mathrm{L}} - \Delta\mu$ and repeat from step 3;
4. Compute $\delta n(\mathbf{v}_i,\mu_{\mathrm{U}})$. Evaluate:
   - if $|\delta n(\mathbf{v}_i,\mu_{\mathrm{U}})| \le \Delta n$, set $\mu_i = \mu_{\mathrm{U}}$ and halt;
   - if $\delta n(\mathbf{v}_i,\mu_{\mathrm{U}}) > \Delta n$ do nothing;
   - if $\delta n(\mathbf{v}_i,\mu_{\mathrm{U}}) < -\Delta n$, shift the upper bound up, $\mu_{\mathrm{U}} \to \mu_{\mathrm{U}} + \Delta\mu$ and repeat from step 4;
5. At this point, the the lower bound underestimates density while the upper bound overestimates. Start iterations:
   a) Compute the midpoint,
   $$\mu_{\mathrm{M}} = \frac{\mu_{\mathrm{L}} + \mu_{\mathrm{U}}}{2}$$
   b) Evaluate:
      - If $|\delta n(\mathbf{v}_i,\mu_{\mathrm{M}})| \le \Delta n$, set $\mu_i = \mu_{\mathrm{M}}$ and halt;
      - If $\delta n(\mathbf{v}_i,\mu_{\mathrm{M}}) > \Delta n$, set $\mu_{\mathrm{U}} = \mu_{\mathrm{M}}$ and repeat from point a);
      - If $\delta n(\mathbf{v}_i,\mu_{\mathrm{M}}) < -\Delta n$, set $\mu_{\mathrm{L}} = \mu_{\mathrm{M}}$ and repeat from point a);

The bottleneck of the procedure is our ability to compute expectation values. This depends on how much we are able to simplify analytically the computations for each phase.

As will turn out in the derivation, chemical potential will be renormalised by interaction. This means that applying self-consistent HF methods to the EHM will produce an effective hamiltonian containing a shift to the chemical potential,

$$\mu \to \mu + F[\mathbf{v}] \equiv \tilde{\mu}[\mathbf{v}]$$

being F some function of the HFPs. The chemical potential μ is an example of Renormalised Model Parameter (RMP).

Note that at step i of the iHF algorithm, our bisection procedure determines $\tilde{\mu}_i$, not μ_i. This consideration is relevant when dealing with ground state energy estimation. Let us abstract from the EHM and iHF, and consider a generic hamiltonian $\hat{H}$. To find the ground state $|\Psi\rangle$ means precisely to solve

$$\frac{\delta}{\delta\langle\Psi|}\langle\Psi|\hat{H}|\Psi\rangle = 0$$

Now, working with a fixed particles number N means we are performing a constrained minimisation (which is, taking the functional derivative) on the composite operator

$$\hat{H} - \mu(\hat{N} - N) \equiv \hat{K} + \mu N$$

being μ the Lagrange multiplier, $\hat{N}$ the number operator and $\hat{K} \equiv \hat{H} - \mu\hat{N}$. Finding the constrained ground state parametrised by a vector $\mathbf{v}$ now means to solve

$$\mathbf{v} \implies |\Psi[\mathbf{v}]\rangle \quad : \quad \begin{cases} \dfrac{\delta}{\delta\langle\Psi|}\langle\Psi|\hat{K} + \mu N|\Psi\rangle \Big|_{|\Psi\rangle = |\Psi[\mathbf{v}]\rangle} = 0 \\ \langle\Psi[\mathbf{v}]|\hat{N}|\Psi[\mathbf{v}]\rangle = N \end{cases}$$

In the first line μN is derived to zero; the constraint is imposed in the second line. Energy is given by:

$$\begin{aligned} \langle\hat{H}\rangle &= \langle\Psi[\mathbf{v}]|\hat{K}|\Psi[\mathbf{v}]\rangle + \mu N \\ &= \Omega[\mathbf{v}] + (\tilde{\mu}[\mathbf{v}] - F[\mathbf{v}])N \end{aligned}$$

being Ω the expectation value (EV) of $\hat{K}$. To correctly compute energy, we need to add $+\mu N$; since we determine $\tilde{\mu}[\mathbf{v}] \neq \mu$, everything needs to be expressed in terms of it. Using this relation, when the algorithm has converged, we can estimate ground-state energy.

2.3.2 The convergence scheme

Let us discuss briefly the convergence criterion of our algorithm and the iteration mechanism. The iHF scheme we design is a sequential loop that runs up to a logical halt condition. In order to specify such condition, we need to define our notion of "convergence". Recall the definition of $\mathrm{A}(\mathbf{v})$, the non-linear map in HFPs space whose component are given by a set of self-consistency equations: we claim to have reached convergence when $\mathbf{v}$ comes "close enough" to a fixed point of the map A, as in Eq. (2.22). Explicitly, we stop whenever the following logic condition is satisfied for each component of $\mathbf{v} = (v_1, v_2, \cdots)$:

$$\forall v_j \quad : \quad |v_j - \mathrm{A}_j(\mathbf{v})| \leq \Delta v_j$$

being $\mathrm{A}(\mathbf{v})$ the map image of the HFP vector $\mathbf{v}$, and A_j its j-th component. The tolerance Δv_j is arbitrary. If just one component of $\mathbf{v}$ is mapped "too far" (with respect to Δv_j) the step is rejected and iterations continue. In this text we deal in with quantities of order 10^{-1}, so convergence values of order $10^{-3} \div 10^{-4}$ will be in general sufficient.

At step i, we check the distance of the two vectors $\mathrm{A}(\mathbf{v}_i)$ and $\mathbf{v}_i$ and if possible, halt. If instead convergence is not reached, we need repeat the cycle with a new initialiser $\mathbf{v}_{i+1}$. A very simple choice would be to set $\mathbf{v}_{i+1} = \mathbf{v}_i$. We do something slightly different:

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}(\mathbf{v}_i) + (1-g)\mathbf{v}_i \tag{2.24}$$

The "mixing parameter" g is a simple but useful knob to tune algorithmic evolution. Its usefulness comes from this observation: even if a fixed point exists, we don't know how easy it is to actually get there following A. Setting $g = 1$, the new initialiser is just the image of the previous initialiser. By moving "slower" ($g < 1$) with an accurate optimisation of the mixing, it is possible to reach convergence in a controlled way.

Let us give some concrete justification about the relevance of controlling g. When it comes to HF iterative schemes of this kind, it can sometimes happen for the algorithm to remain stuck in infinite loops. An example of this behaviour is given in Sec. 4.5.3. A typical situation is the following: during algorithm evolution we reach a couple of points $\mathbf{v}^{(a)}$, $\mathbf{v}^{(b)}$ such that

$$\dots \xrightarrow{\text{A},g} \mathbf{v}^{(a)} \xrightarrow{\text{A},g} \mathbf{v}^{(b)} \xrightarrow{\text{A},g} \mathbf{v}^{(a)} \xrightarrow{\text{A},g} \mathbf{v}^{(b)} \xrightarrow{\text{A},g} \dots$$

With the notation "$\xrightarrow{\text{A},g}$" we intend one algorithmic step at mixing parameter g. We are stuck forever in this loop. Now, to predict if couples like $(\mathbf{v}^{(a)}, \mathbf{v}^{(b)})$ exist, how they depend on g, where they are located in HFPs space... is impossible at this level. Thus there is no immediate way to avoid these loops *a priori*. One workaround we found to work in most cases, however, was to just make g smaller. We gained empirical evidence that to "slow down" in HFPs space makes numeric instabilities less frequent. The trade-off, however, is that if we move too slow, runtime could get very large. Note that inaccurate setup of the algorithm could lead to resources-intensive runs also in "ordinary" situations, when we are not stuck in a loop.

To keep the number of iterations under control, we also defined an upper bound p to the total number of steps. If after p steps we algorithm failed to converge, it halts. In that case, $\mathbf{v}$ is saved as a "`NaN`" (Not a number) vector and we move to another simulation. Introducing this simple rule, runtime is controlled and we gain an easy method to identify the regions of parameter space requiring more care – just read from the raw data where `NaN`s have been saved and repeat simulations fine-tuning the parameters. This basically means to lower g and increase p. These considerations end this technical introductory chapter on iHF methods.

Chapter 3

Phase transitions and SSB in EHM

In this chapter we discuss the symmetries of the EHM, and describe the phases we will investigate in next chapter based on which symmetry they break. Then we connect this discussion with the HF method by deriving a set of selection rules on the HF approximation of the EHM hamiltonian for each phase.

This chapter serves as groundwork for the solution of the self-consistency problem posed in Eq. (2.19) for each phase. We start by enumerating and demonstrating the symmetries of the EHM.

3.1 The EHM symmetries

In this section we present the symmetries of the extended Hubbard model introduced in Eq. (1.2)

$$\hat{H}_{\text{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\hat{H}_t} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\hat{H}_U} \underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{n}_{i\sigma}\hat{n}_{j\sigma'}}_{\hat{H}_V}$$

The symmetries of the model can be divided in two groups: the ones related to the geometry of the square lattice, and those related to the algebraic properties of $\hat{H}_{\text{EHM}}$. We start with the lattice symmetries.

3.1.1 Lattice symmetries

We study the EHM on the planar square lattice, with one orbital per site. We indicate with L_x the number of sites in direction x, and L_y in direction y. The hamiltonian of Eq. (1.2) respects three fundamental lattice symmetries:

1. Discrete translations symmetry in both the x and y directions. We refer to the groups of one-site translations as as $\mathrm{T}_1^{(x)}$, $\mathrm{T}_1^{(x)}$. It holds:

   $$\mathrm{T}_n^{(\ell)} \supset \mathrm{T}_{kn}^{(\ell)} \qquad k \in \mathbb{N} \tag{3.1}$$

   A shift in direction ℓ by kn sites can be decomposed in k shift by n sites. Invariance by n-sites shifts, thus, implies invariance also for kn-sites shifts.

2. Discrete $\pi/2$ rotations symmetry. This symmetry is described by the four-fold cyclic point group C_4, containing rotations by angle $\pi/2, \pi, 3\pi/2$ and identity. A similar property as for the translations holds for cyclic point groups,

   $$\mathrm{C}_n \supset \mathrm{C}_{n/k} \qquad k, n/k \in \mathbb{N} \tag{3.2}$$

   Invariance for rotations of angle $2\pi/n$ implies invariance for rotations of angle $k \times 2\pi/n$.

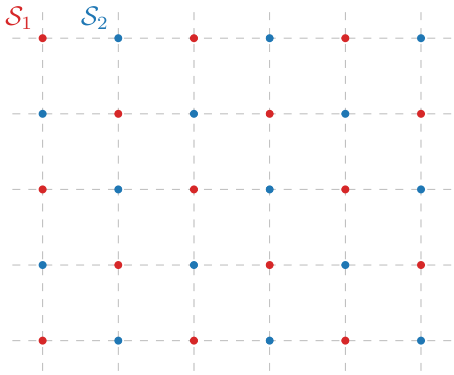


Figure 3.1 | Square lattice $\mathcal{L}$ bipartition in two sublattices $\mathcal{S}_1$ (red), $\mathcal{S}_2$ (blue).

3. Inversion symmetry. We define I_2 as the inversion group in two-dimensional space, containing the reflection operator and identity. This point may appear redundant, because

$$\mathrm{I}_2 \equiv \mathrm{C}_2 \tag{3.3}$$

In 2D space, to rotate by π around the origin and to mirror the position of a point are the same operation.

The planar square lattice is bipartite. Let us give a definition. Consider a graph $\mathcal{L}$ and the set of its vertices $V(\mathcal{L}) = \{v_i\}$. Let $E(\mathcal{L})$ be the set of graph edges,

$$E(\mathcal{L}) = \{(v, w) \mid v, w \in V(\mathcal{L}) \wedge v,\, w \text{ are connected}\}$$

The graph is said to be bipartite if two sub-graphs $\mathcal{S}_1$, $\mathcal{S}_2$ exist such that:

- The vertices of $\mathcal{L}$ split in the two sub-graphs,

$$V(\mathcal{L}) = V(\mathcal{S}_1) \cup V(\mathcal{S}_2)$$

- Vertices of one sub-graph are linked only to vertices of the other sub-graph. In other words

$$v, w \in \mathcal{S}_i \quad \Longrightarrow \quad (v, w) \notin E(\mathcal{L})$$

The planar square lattice is bipartite: it can be divided in two square lattices, as represented in Fig. 3.1. Two sites are "connected" if linked by a non-zero hopping operator. Which is, if they are NNs. Note that one-site translations leave the hamiltonian invariant but exchange the two sublattices, while two-sites translations don't.

3.1.2 Spin and charge symmetries

The conventional HM shows a global U(2) symmetry, extended to $\mathrm{SU}(2) \times \mathrm{SU}(2)/\mathbb{Z}_2 = \mathrm{SO}(4)$ symmetry at half-filling on a bipartite lattice [14], such as the square lattice. For the EHM, we limit to prove that we can identify a $\mathrm{U}(2) \subset \mathrm{SO}(4)$ global symmetry.

Gauge symmetry. The EHM is symmetric under the global transformation

$$\hat{c}_{i\sigma} \to e^{-i\eta} \hat{c}_{i\sigma} \qquad \hat{c}^\dagger_{i\sigma} \to e^{i\eta} \hat{c}^\dagger_{i\sigma} \qquad \eta \in \mathbb{R} \tag{3.4}$$

To prove this, it suffices to note that

$$\hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \quad \to \quad e^{i\eta} \hat{c}^\dagger_{i\sigma} e^{-i\eta} \hat{c}_{j\sigma'} = \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \tag{3.5}$$

Any product of fermionic operators with an equal number of annihilations and creations is gauge-symmetric. Hence the kinetic part $\hat{H}_t$ and the interactions $\hat{H}_U$, $\hat{H}_V$ are invariant.

We observe that the on-site density is gauge invariant,

$$\hat{n}_{i\sigma} \to \hat{n}_{i\sigma} \tag{3.6}$$

It follows that the total number of particles (charge)

$$\hat{N} = \sum_{i\sigma} \hat{c}^{\dagger}_{i\sigma}\hat{c}_{i\sigma}$$

is a good quantum number.

Gauge symmetry is described by the transformation of Eq. (3.4), which is a pure phase shift by η. The gauge group is $\mathrm{U}^{\mathrm{c}}(1)$, with a superscript “c” indicating “charge”.

Global spin symmetry The EHM is described by three operators: the kinetic operator $\hat{H}_t$, independent of σ; the local repulsion $\hat{H}_U$ between opposite-spin densities; the non-local attraction $\hat{H}_V$, independent of σ. If we rotate all spins by an equal amount, the hamiltonian does not change. An identical observation holds for the number operator, $\hat{N}$. Thus $\hat{H} - \mu\hat{N}$ must be exhibit a global SU(2) symmetry, related to the rotation in 3D space of all spins altogether.

Let us discuss this rigorously. We define the spinors $\hat{\Psi}_i$,

$$\hat{\Psi}_i \equiv \begin{bmatrix} \hat{c}_{i\uparrow} \\ \hat{c}_{i\downarrow} \end{bmatrix} \tag{3.7}$$

and the Pauli matrices

$$\tau^x \equiv \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \qquad \tau^y \equiv \begin{bmatrix} & -i \\ i & \end{bmatrix} \qquad \tau^z \equiv \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}$$

The local spin operator are defined as

$$\hat{S}^{\alpha}_i \equiv \frac{1}{2}\hat{\Psi}^{\dagger}_i \tau^{\alpha} \hat{\Psi}_i \qquad \left[\hat{S}^{\alpha}_j, \hat{S}^{\beta}_j\right] = i\sum_{\gamma} \epsilon_{\alpha\beta\gamma}\hat{S}^{\gamma}_j \tag{3.8}$$

with $\epsilon_{\alpha\beta\gamma}$ the Levi-Civita tensor. The $\mathfrak{su}(2)$ spin algebra is actually respected:

Proof 3.1 (Spin operators respect the $\mathfrak{su}(2)$ algebra).

To prove that $\mathfrak{su}(2)$ algebra is respected, we must compute all possible commutators. This involves many very similar calculations. For the sake of conciseness, we sketch the proof for $\alpha = x$, $\beta = y$:

$$\begin{aligned}
\left[\hat{S}^x_j, \hat{S}^y_j\right] &= \frac{1}{4}\left[\hat{\Psi}^{\dagger}_j \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \hat{\Psi}_j, \hat{\Psi}^{\dagger}_j \begin{bmatrix} & -i \\ i & \end{bmatrix} \hat{\Psi}_j\right] \\
&= \frac{1}{4}\left[\hat{c}^{\dagger}_{j\uparrow}\hat{c}_{j\downarrow} + \hat{c}^{\dagger}_{j\downarrow}\hat{c}_{j\uparrow}, -i\hat{c}^{\dagger}_{j\uparrow}\hat{c}_{j\downarrow} + i\hat{c}^{\dagger}_{j\downarrow}\hat{c}_{j\uparrow}\right] \\
&= \frac{i}{4}\left[\hat{c}^{\dagger}_{j\uparrow}\hat{c}_{j\downarrow}, \hat{c}^{\dagger}_{j\downarrow}\hat{c}_{j\uparrow}\right] - \frac{i}{4}\left[\hat{c}^{\dagger}_{j\downarrow}\hat{c}_{j\uparrow}, \hat{c}^{\dagger}_{j\uparrow}\hat{c}_{j\downarrow}\right] \\
&= \frac{i}{2}\left[\hat{c}^{\dagger}_{j\uparrow}\hat{c}_{j\downarrow}, \hat{c}^{\dagger}_{j\downarrow}\hat{c}_{j\uparrow}\right] \\
&= \frac{i}{2}\left(\hat{c}^{\dagger}_{j\uparrow}\hat{c}_{j\downarrow}\hat{c}^{\dagger}_{j\downarrow}\hat{c}_{j\uparrow} - \hat{c}^{\dagger}_{j\downarrow}\hat{c}_{j\uparrow}\hat{c}^{\dagger}_{j\uparrow}\hat{c}_{j\downarrow}\right) \\
&= \frac{i}{2}\left(-\hat{c}^{\dagger}_{j\uparrow}\hat{c}^{\dagger}_{j\downarrow}\hat{c}_{j\downarrow}\hat{c}_{j\uparrow} + \hat{c}^{\dagger}_{j\uparrow}\hat{c}_{j\uparrow} + \hat{c}^{\dagger}_{j\downarrow}\hat{c}^{\dagger}_{j\uparrow}\hat{c}_{j\uparrow}\hat{c}_{j\downarrow} - \hat{c}^{\dagger}_{j\downarrow}\hat{c}_{j\downarrow}\right) \\
&= \frac{i}{2}\left(\hat{c}^{\dagger}_{j\uparrow}\hat{c}_{j\uparrow} - \hat{c}^{\dagger}_{j\downarrow}\hat{c}_{j\downarrow}\right) \\
&= i\hat{S}^z_j
\end{aligned}$$

We used Fermi anticommutation rules, $\{\hat{c}_{i\sigma}, \hat{c}^{\dagger}_{j\sigma'}\} = \delta_{ij}\delta_{\sigma\sigma'}$. Identical derivations can be carried out for all other commutators, giving in the end the $\mathfrak{su}(2)$ spin algebra.

We define the on-site raising and lowering operators:

$$\hat{S}_j^+ \equiv \frac{\hat{S}_j^x + i\hat{S}_j^y}{2} \qquad \hat{S}_j^- \equiv \frac{\hat{S}_j^x - i\hat{S}_j^y}{2}$$

and the global spin operators

$$\hat{S}^\alpha \equiv \sum_{i\in\mathcal{L}} \hat{S}_i^\alpha \qquad \left[\hat{S}^\alpha, \hat{S}^\beta\right] = i\sum_\gamma \epsilon_{\alpha\beta\gamma}\hat{S}^\gamma$$

being $\mathcal{L}$ the lattice. Explicitly,

$$\hat{S}^z \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} (\hat{n}_{i\uparrow} - \hat{n}_{i\downarrow}) \qquad \hat{S}^+ \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow} \qquad \hat{S}^- \equiv \left(\hat{S}^+\right)^\dagger$$

The EHM hamiltonian and the number operator commutate with the global spin operator. For the kinetic part $\hat{H}_t$ and the local repulsion $\hat{H}_U$ this is a well known result [14, 27],

$$\left[\hat{H}_t + \hat{H}_U, \hat{\mathbf{S}}\right] = 0 \qquad \left[\hat{N}, \hat{\mathbf{S}}\right] = 0$$

For the non-local attraction, since it depends solely on total densities on neighbouring sites, spin-rotational invariance must hold,

$$\left[\hat{H}_V, \hat{\mathbf{S}}\right] = 0 \tag{3.9}$$

Let us give a formal proof.

Proof 3.2 (The non-local attraction $\hat{H}_V$ is symmetric under global spin rotations.).

The proof is organised through a sequence of results:

1. The non-local attraction can be written as:

$$\begin{aligned}\hat{H}_V &= V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'} \hat{n}_{i\sigma}\hat{n}_{j\sigma'} \\ &= V\sum_{\langle ij\rangle} \hat{\Psi}_i^\dagger\hat{\Psi}_i\hat{\Psi}_j^\dagger\hat{\Psi}_j\end{aligned} \tag{3.10}$$

   having we used $\hat{\Psi}_i^\dagger\hat{\Psi}_i = \hat{c}_{i\uparrow}^\dagger\hat{c}_{i\uparrow} + \hat{c}_{i\downarrow}^\dagger\hat{c}_{i\downarrow}$.

2. Let now $\hat{R}(\boldsymbol{\theta})$ be any global spin rotation,

$$\hat{R}(\boldsymbol{\theta}) \equiv \exp\left(-i\boldsymbol{\theta}\cdot\hat{\mathbf{S}}\right) \quad \text{where} \quad \boldsymbol{\theta} = (\theta_x, \theta_y, \theta_z)$$

   Spinors transform as:

$$\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger = \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i$$

   with $\mathcal{R}(\boldsymbol{\theta}) \in \mathrm{SU}(2)$ the unitary matrix associated to said rotation,

$$\mathcal{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right) \quad \text{where} \quad \boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z) \tag{3.11}$$

   Let us prove this. Let $\hat{A} = i\boldsymbol{\theta}\cdot\hat{\mathbf{S}}$ e $\hat{B} = \hat{\Psi}_i$. The following identity, a consequence of Baker-Campbell-Housdorff formula, holds:

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + \left[\hat{A}, \hat{B}\right] + \frac{1}{2!}\left[\hat{A}, \left[\hat{A}, \hat{B}\right]\right] + \frac{1}{3!}\left[\hat{A}, \left[\hat{A}, \left[\hat{A}, \hat{B}\right]\right]\right] + \cdots$$

The first order of the expansion gives:

$$
\begin{aligned}
[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\hat{\Psi}_i] &= i\sum_\alpha \theta_\alpha[\hat{S}^\alpha,\hat{\Psi}_i] \\
&= -\frac{i}{2}\sum_\alpha \theta_\alpha \tau^\alpha \hat{\Psi}_i \\
&= -\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\hat{\Psi}_i
\end{aligned}
$$

We omitted some purely algebraic passages. The second order:

$$
\begin{aligned}
\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\hat{\Psi}_i\Big]\Big] &= \Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},-\frac{i}{2}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})\hat{\Psi}_i\Big] \\
&= -\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\Psi_i\Big] \\
&= \frac{1}{4}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})^2\hat{\Psi}_i
\end{aligned}
$$

and so on. Collecting all terms,

$$
\begin{aligned}
\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger &= \hat{\Psi}_i - \frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\hat{\Psi}_i + \frac{1}{2!\times 4}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})^2\hat{\Psi}_i + \cdots \\
&= \sum_{k=1}^{\infty}\frac{1}{k!}\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right)^k\hat{\Psi}_i \\
&= \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i
\end{aligned}
$$

In the last line, we identified the series as the Taylor expansion of the exponential, recognising the expression for $\mathcal{R}(\boldsymbol{\theta})$ given in Eq. (3.11).

3. Then, the group acts on the product $\hat{\Psi}_i^\dagger\hat{\Psi}_i$ as

$$
\begin{aligned}
\hat{R}\hat{\Psi}_i^\dagger\hat{\Psi}_i\hat{R}^\dagger &= \hat{R}\hat{\Psi}_i^\dagger\hat{R}^\dagger\hat{R}\hat{\Psi}_i\hat{R}^\dagger \\
&= \hat{\Psi}_i^\dagger\mathcal{R}^\dagger\mathcal{R}\hat{\Psi}_i \\
&= \hat{\Psi}_i^\dagger\hat{\Psi}_i
\end{aligned}
$$

being $\mathcal{R}\in \mathrm{SU}(2)$. Recalling Eqns. (3.10), we conclude that $\hat{H}_V$ is SU(2) symmetric – which means, it does not change under global spin rotations. This fact implies Eq. (3.9).

We observe that the above proof can be used to show that the on-site number operator is spin-invariant,

$$
\hat{n}_{i\sigma} = \hat{\Psi}_i^\dagger\hat{\Psi}_i \quad\Longrightarrow\quad \Big[\hat{n}_{i\sigma},\hat{\mathbf{S}}\Big] = 0 \tag{3.12}
$$

The EHM is symmetric under global spin rotations. Specifically, both $\hat{H}_U$ and $\hat{H}_V$ are symmetric also under local spin rotations, meaning that we could rotate the spin at each site by a different amount and get the same interaction. It is the kinetic operator that makes the EHM globally (and not locally) SU(2) symmetric.

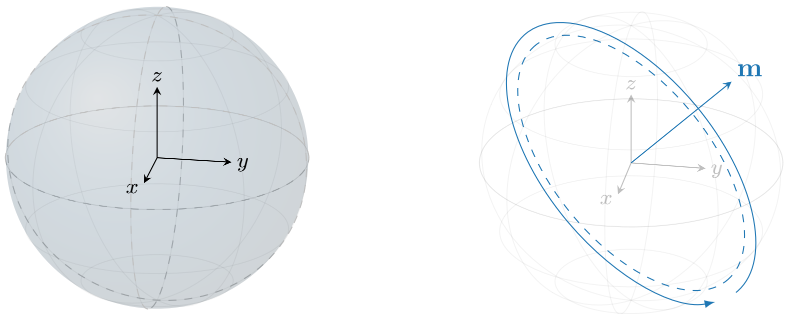
The $\mathrm{U^c}(1)$ gauge symmetry and the SU(2) spin rotations symmetry can be combined,

$$
\mathrm{U^c}(1)\otimes\mathrm{SU}(2) = \mathrm{U}(2)
$$

We will refer to U(2) symmetry to indicate full charge-spin conservation. Finally, note the group of 3D rotations contains the subgroup of 2D rotations around a given axis z,

$$
\mathrm{U}^z(1)\subset\mathrm{SU}(2)
$$

where the z superscript serves to distinguish from the gauge group. The z axis does not need to be identified with the "real" z axis, perpendicular with the lattice plane. Any given direction works.



a | SU(2) rotations domain before SSB.

b | U(1) rotations domain after SSB.

Figure 3.2 | Graphic representation of the concept of $\mathrm{SU}(2) \to \mathrm{U}^z(1)$ SSB. "Before" SSB, we have full symmetry under any possible rotation (solid color sphere, left panel). "After" SSB we have symmetry only for rotations around some vector $\mathbf{m}$ (dashed tilted circumference, right panel).

Inverting our logic, we can build the SU(2) generators, starting from $\hat{S}^z$, after we have chosen the direction z. In terms of commutation relations, a given operator can respect $\mathrm{U}^z(1)$ but not SU(2): in that case, it will commutate with $\hat{S}^z$, and not the other spin operators. A graphic rendition of this SSB is given in Fig. 3.2, representing for a given vector $\mathbf{m}$ the passage from full SU(2) rotational symmetry around any direction to reduced $\mathrm{U}^z(1)$ rotational symmetry around just one direction.

3.1.3 Particle-hole symmetry at half-filling

The half-filled EHM on the square lattice exhibits particle-hole symmetry (PHS). The presence of this additional symmetry is a consequence of the fact that the square lattice is bipartite. The generic unitary particle-hole (PH) transformation is given by

$$\mathrm{P} \quad : \quad \hat{c}_{i\sigma} \quad \to \quad e^{i\eta_{i\sigma}} \hat{h}^\dagger_{i\sigma} \tag{3.13}$$

being in general η_i a phase dependent on the fermionic quantum state $(i\sigma)$. Inserting the above transformation in the EHM, we get

$$\mathrm{P}\{\hat{H}\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})} \hat{h}_{i\sigma} \hat{h}^\dagger_{j\sigma} + U \sum_i \hat{h}_{i\uparrow} \hat{h}_{i\downarrow} \hat{h}^\dagger_{i\downarrow} \hat{h}^\dagger_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}^\dagger_{j\sigma'} \hat{h}^\dagger_{i\sigma}$$

For the two interactions $\hat{H}_U$, $\hat{H}_V$ the phase factors cancelled out. For a specific choice of the η_i phases,

$$\mathrm{P} \quad : \quad \begin{cases} \hat{c}_{i\sigma} \to \hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_1 \\ \hat{c}_{i\sigma} \to -\hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_2 \end{cases} \tag{3.14}$$

the above hamiltonian is mapped onto the EHM of (1.2),

$$\mathrm{P}\{\hat{H}\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{h}^\dagger_{i\sigma} \hat{h}_{j\sigma} + U \sum_{i \in \mathcal{L}} \hat{h}^\dagger_{i\uparrow} \hat{h}^\dagger_{i\downarrow} \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'} \hat{h}_{i\sigma}$$

The equality $\mathrm{P}\{\hat{H}\} = \hat{H}$ expresses PHS, which holds as long as we work at half-filling. Let us give a proof.

Proof 3.3 (PHS at half filling, with η_i given by Eq. (3.14)).

This proof is an extension of the argument carried out for the conventional HM [14]. For the hole operators, Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned}\{\hat{h}_{i\sigma}, \hat{h}^\dagger_{j\sigma'}\} &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\{\hat{c}^\dagger_{i\sigma}, \hat{c}_{j\sigma'}\}\\ &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\delta_{ij}\delta_{\sigma\sigma'}\\ &= \delta_{ij}\delta_{\sigma\sigma'}\end{aligned}$$

The on-site density operator is given by:

$$\hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma} = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site i with spin σ. It follows, for quartic interactions

$$\begin{aligned}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} &= -\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\\ &= \hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'}\\ &= -\hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'}\\ &= \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'}\\ &= -\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'}\\ &= \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \qquad (3.15)\end{aligned}$$

The first term of the last passage is mirrored with respect to the starting term. In next paragraphs we derive the constraint of Eq. (3.14) and the requirement to work at half-filling.

Kinetic term. Consider the kinetic term,

$$\hat{H}_t = -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}$$

We apply the PH transform of Eq. (3.13)

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma}e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma}$$

Anti-commutating the two fermionic operators, and exchanging site labels $i \leftrightarrow j$, we get:

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma}e^{i(\eta_{i\sigma}+\eta_{j\sigma}+\pi)}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma}$$

To obtain $\mathrm{P}\{\hat{H}_t\} \overset{!}{=} \hat{H}_t$, a necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} \overset{!}{=} (2k-1)\pi \quad \text{with} \quad k \in \mathbb{Z} \qquad (3.16)$$

valid for NN sites. This requirement on the phases η_i gives Eq. (3.14).

Local repulsion. Substitute Eq. (3.15) in $\hat{H}_U$,

$$\begin{aligned}\mathrm{P}\{\hat{H}_U\} &= U\sum_{i\in\mathcal{L}}\hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^\dagger_{i\downarrow}\hat{h}^\dagger_{i\uparrow}\\ &= U\sum_{i\in\mathcal{L}}\hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U\sum_{i\in\mathcal{L}}(\hat{n}_{i\uparrow}+\hat{n}_{i\downarrow}) - U\sum_{i\in\mathcal{L}}(1)\\ &= U\sum_{i}\hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U(\hat{N} - L_xL_y)\end{aligned}$$

Symmetry	Operations
$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$	One-site discrete translations separately in both directions
C_4	Four-fold point group (real $\pi/2$ angle rotations)
$\mathrm{U}^{c}(1)$	Charge-conserving global phase shifts
$\mathrm{SU}(2)$	Three-dimensional rotations in global spin space

Table 3.1 Model symmetries and relative group operations. Note that $\mathrm{U}^{c}(1)$ represents the gauge symmetry linked to charge conservation.

At half filling, $\langle \hat{N} \rangle = L_x L_y$. The unitary map of Eq. (3.13), independently of the condition of Eq. (3.16), leaves the hamiltonian identical at half-filling.

Non-local attraction. We proceed identically for $\hat{H}_V$,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_V\} &= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}^{\dagger}_{j\sigma'} \hat{h}^{\dagger}_{i\sigma} \\
&= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^{\dagger}_{i\sigma} \hat{h}^{\dagger}_{j\sigma'} \hat{h}_{j\sigma'} \hat{h}_{i\sigma} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) + V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (1) \\
&= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^{\dagger}_{i\sigma} \hat{h}^{\dagger}_{j\sigma'} \hat{h}_{j\sigma'} \hat{h}_{i\sigma} - 2zV(\hat{N} - L_x L_y)
\end{aligned}$$

with $z = 4$ the lattice coordination number. In the last line, we used that on the square lattice

$$\begin{aligned}
\sum_{\langle ij \rangle} (1) &= \sum_{i \in \mathcal{L}/2} z \\
&= \frac{L_x L_y}{2} \times z
\end{aligned} \tag{3.17}$$

As before, the non-local attraction is particle-hole symmetric at half-filling. We conclude that the entire EHM exhibits PHS at half filling, if P is defined as in Eq. (3.14).

As for the conventional HM [14], this result can be generalised to any bipartite lattice. In turn, if a lattice is not bipartite, PHS is not present. To summarise, we report in Tab. 3.1 all identified symmetries.

3.2 The HF method and symmetry breaking

In Quantum Mechanics, a phase transition is described by the breaking of some natural symmetry of the hamiltonian. In this thesis we investigate three phases:

- Normal phase: no broken symmetry.
- Antiferromagnetic phase: simultaneous $\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^{z}(1)$ and $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}$. We study the so-called "commensurate antiferromagnetism".
- Superconducting phase: simultaneous $\mathrm{U}^{c}(1) \overset{\mathrm{SSB}}{\to} 0$ and $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$. We investigate anisotropic superconductivity and breaking of planar rotational symmetry, observed in unconventional superconductors [23].

In the subsections that follow we discuss how to implement these SSBs in the context of the HF method. Indeed, we investigate each of these phases by approximating the EHM hamiltonian

$$\hat{H} \overset{\mathrm{HF}}{\simeq} \hat{H}_{\mathrm{HF}}$$

with an operator $\hat{H}_{\mathrm{HF}}$ that respects the symmetries of the given phase. Hence, starting by the general decomposition of the two interactions $\hat{H}_U$ and $\hat{H}_V$, given respectively in Eqns. (2.20) and (2.21), we discuss the symmetry properties of each sector.

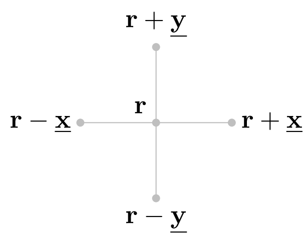


Figure 3.3 | Schematic representation of the four NNs of a given site i for a planar square lattice.

In next subsections, we switch from the site index $i \in \mathcal{L}$ to the vector notation $\mathbf{r} \in \mathbb{N}^2$, indicating the site 2D position on the lattice

$$i \quad \rightarrow \quad \mathbf{r} = (x, y)$$

with $x, y \in \mathbb{N}$. The lattice versors are:

$$\underline{\mathbf{x}} \equiv (1,0) \qquad \underline{\mathbf{y}} \equiv (0,1)$$

as in Fig. 3.3.

3.2.1 Local repulsion

The local repulsion $\hat{H}_U$ is given by:

$$\begin{aligned}\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow} \\ &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\end{aligned}$$

Applying Wick's theorem and using the denominations of Eq. (2.1) we have:

$$\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle = \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} \tag{3.18}$$

where the equality holds exactly as long as we perform the averaging operation on the subspace of gaussian states. We now discuss the above EVs separately, obtaining selection rules for each phase.

Hartree term. The Hartree part of the above decomposition is a product of density EVs,

$$\langle \hat{n}_{\mathbf{r}\uparrow}\rangle\langle \hat{n}_{\mathbf{r}\downarrow}\rangle$$

From Eq. (3.6), (3.12) we know that single-site number operators are invariant under the entire group U(2). Moreover, shifting or rotating the lattice leaves density operators unchanged. Thus no selection rule applies, and the above EVs are in principle non-zero.

Fock terms. For the Fock part, notice that

$$\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow} = \hat{S}^+_{\mathbf{r}}$$

being $S^+_{\mathbf{r}}$ the local raising operator. From $\mathfrak{su}(2)$ algebra we have:

$$\left[\hat{S}^+_{\mathbf{r}}, \hat{S}^\alpha\right] \neq 0 \qquad \alpha \in \{x, y, z\}$$

If the ground-state is invariant under global spin rotations, $\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}\rangle$ must be zero. instead, gauge symmetry gives no selection rules, due to Eq. (3.5).

Term	Normal phase	Antiferromagnet	Superconductor
Hartree			
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	

Table 3.2 Summary of selection rules for the Wick's decomposition of Eq. (3.18) of the local repulsion $\hat{H}_U$. An empty entry indicates no selection rule.

Cooper terms. The Cooper part can be non-zero only if charge is not conserved. In terms of the gauge symmetry, it suffices to apply the map of Eq. (3.4) to $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$

$$\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} \quad \rightarrow \quad e^{2i\eta}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$$

to conclude that this operator is in general not gauge invariant. It is instead SU(2)-symmetric. Let us provide a proof:

Proof 3.4 (Local Cooper terms are SU(2)-symmetric).

We must compute the commutators of $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$ with $\hat{S}^z$, $\hat{S}^+$:

$$\begin{aligned}
\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow})\right] \\
&= \frac{1}{2}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] \\
&= \frac{1}{2}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}\right]\hat{c}^\dagger_{\mathbf{r}\downarrow} - \frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\left[\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}\right]
\end{aligned}$$

Since:

$$\begin{aligned}
\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} - \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\sigma} \\
&= -\hat{c}^\dagger_{\mathbf{r}\sigma}\left(1 - \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\right) \\
&= -\hat{c}^\dagger_{\mathbf{r}\sigma}
\end{aligned} \tag{3.19}$$

we have:

$$\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^z\right] = -\frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} + \frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} = 0$$

Hence, $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$ is $\mathrm{U}^z(1)$-symmetric.

Moreover, considering the raising operator:

$$\begin{aligned}
\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^+\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}\hat{c}^\dagger_{\mathbf{r}'\uparrow}\hat{c}_{\mathbf{r}'\downarrow}\right] \\
&= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}\right] \\
&= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}, \hat{c}^\dagger_{\mathbf{r}\uparrow}\right]\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} - \left(\hat{c}^\dagger_{\mathbf{r}\uparrow}\right)^2\left[\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{c}_{\mathbf{r}\downarrow}\right] \\
&= 0
\end{aligned}$$

where we used the facts that every operator commutates with itself, and a squared creation operator is null. These commutation relations, together, prove that Cooper EVs are legitimately non-zero for a SU(2)-preserving phase.

In Tab. 3.2 we collect the selection rules for the Hartree, Fock and Cooper terms of Eq. (3.18), based on the symmetry specifications given for each phase in above paragraphs. Note that Fock-like EVs are prohibited for all phases.

3.2.2 Non-local attraction

The non-local attraction $\hat{H}_V$ is given by:

$$\begin{aligned}\hat{H}_V &= -V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma,\sigma'} \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\sigma'} \\ &= V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma,\sigma'} \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}'\sigma'}\hat{c}_{\mathbf{r}'\sigma'}\hat{c}_{\mathbf{r}\sigma}\end{aligned} \tag{3.20}$$

Let

$$\hat{H}_V^{\sigma\sigma'} \equiv -V \sum_{\langle \mathbf{r},\mathbf{r}'\rangle} \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}'\sigma'}$$

The interaction decomposes in two "spin sectors"

$$\begin{aligned}\hat{H}_V &= \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite spin}} \\ &\equiv \hat{H}^{(\text{s.s.})} + \hat{H}^{(\text{o.s.})}\end{aligned} \tag{3.21}$$

where we further separated in s.s. and o.s. operators.

The operator $\hat{H}^{(\text{s.s.})}$ can be written as follows:

$$\begin{aligned}\hat{H}_V^{(\text{s.s.})} &\equiv \hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow} \\ &= -V \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma}\right)\end{aligned} \tag{3.22}$$

where the sum over $\mathcal{L}/2$, in last line, means that we are summing over just one sublattice (it does not matter which one). Applying Wick's theorem in the subspace of gaussian states, we get:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle \\ &= \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle\langle \hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle}_{\text{Cooper}}\end{aligned} \tag{3.23}$$

Let us demonstrate Eq. (3.22).

Proof 3.5 (Proof of Eq. (3.22)).

Let $f(\mathbf{r},\mathbf{r}')$ be a function of two variables on the square lattice. Summing over NNs, we get the identity

$$\sum_{\langle \mathbf{r}\mathbf{r}'\rangle} f(\mathbf{r},\mathbf{r}') = \sum_{\mathbf{r}\mathbf{r}'} f(\mathbf{r},\mathbf{r}')\left(\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right)$$

where we used the notation of Fig. 3.3. The above sum is identically written as:

$$\sum_{\langle \mathbf{r}\mathbf{r}'\rangle} f(\mathbf{r},\mathbf{r}') = \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left[f(\mathbf{r},\mathbf{r}+\boldsymbol{\delta}) + f(\mathbf{r},\mathbf{r}-\boldsymbol{\delta})\right] \tag{3.24}$$

The symbol "$\mathcal{L}/2$" indicates one of the two sublattices $\mathcal{S}_1$, $\mathcal{S}_2$ represented in Fig. 3.1. Inserting Eq. (3.24) in Eq. (3.20), we obtain:

$$\hat{H}_V = -V \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \sum_{\sigma,\sigma'} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right) \tag{3.25}$$

By taking the s.s. sector of the above equation, namely $\sigma = \sigma'$, we obtain Eq. (3.22).

The operator $\hat{H}^{(\text{o.s.})}$ can be written as follows:

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} &\equiv \hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow} \\ &= -V \sum_{\mathbf{r}\in\mathcal{L}} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right)\end{aligned} \tag{3.26}$$

From Wick's theorem:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &= \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}}\end{aligned} \tag{3.27}$$

Again, equality holds in the subspace of gaussian states. Let us prove Eq. (3.26).

Proof 3.6 (Proof of Eq. (3.26)).

Consider Eq. (3.25), restricting to $\sigma = \uparrow$, $\sigma' = \downarrow$ and setting $\mathcal{L}/2 = \mathcal{S}_1$. We obtain

$$\hat{H}_V^{\uparrow\downarrow} = -V \sum_{\mathbf{r}\in\mathcal{S}_1}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} (\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow})$$

Note that spin $\uparrow$ operators act on the sublattice $\mathcal{S}_1$; spin $\downarrow$ operators act on the sublattice $\mathcal{S}_2$. If instead, from Eq. (3.25), we restrict to $\sigma = \downarrow$, $\sigma' = \uparrow$ and set $\mathcal{L}/2 = \mathcal{S}_2$, we get:

$$\hat{H}_V^{\downarrow\uparrow} = -V \sum_{\mathbf{r}\in\mathcal{S}_2}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} (\hat{n}_{\mathbf{r}\downarrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\uparrow})$$

We observe that if $\mathbf{r} \in \mathcal{S}_2$, all its NNs belong to $\mathcal{S}_1$. Again, spin $\uparrow$ operators act on the sublattice $\mathcal{S}_1$; spin $\downarrow$ operators act on the sublattice $\mathcal{S}_2$.

Therefore, summing $\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}$, we obtain

$$\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow} = -V \sum_{\mathbf{r}\in\mathcal{L}}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} (\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow})$$

which coincides with Eq. (3.26).

The separation of $\hat{H}_V$ in o.s. and s.s. sector is necessary in order to derive a set of selection rules for the Wick's contractions of Eqns. (3.23), (3.27). This is done in next paragraphs.

Hartree terms. As was for the local repulsion $\hat{H}_U$, the non-local Hartree terms of Eqns. (3.26), (3.22) are not subject to selection rules for any symmetry of Tab. 3.1. Thus they are allowed in both sectors.

Fock terms. The Fock EVs from the two sectors behave differently. Let us compute the following generic commutation relations:

$$\begin{aligned}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow})\right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right]\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\hat{c}^\dagger_{\mathbf{r}\sigma}[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\right)\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\end{aligned} \tag{3.28}$$

where in last passage we used Eq. (3.19) and its conjugate version,

$$[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma}] = \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}$$

Now, if $\sigma = \sigma'$ the last passage of Eq. (3.28) is zero; otherwise is not. This proves that the o.s. Fock EVs are non-zero only for a phase that completely breaks SU(2); instead, in the s.s. sector we can have non-zero EVs for a phase that reduces SU(2) to $\mathrm{U}^z(1)$. [Commutation with $\hat{S}^+$?]

Sector	Term	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree			
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	
s.s.	Hartree			
	Fock			
	Cooper	Prohibited	Prohibited	Prohibited

Table 3.3 Summary of selection rules for the Wick's decomposition of Eqns. (3.27), (3.23) of the non-local attraction $\hat{H}_V$. An empty entry indicates no selection rule.

Cooper terms. Cooper terms are not gauge invariant. The proof is analogous to the one given for $\hat{H}_U$. Instead, the o.s. and s.s. spin sectors behave differently under global spin rotations. With an identical procedure as for Eq. (3.28), we get:

$$\begin{aligned}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow}-\hat{n}_{\mathbf{r}'\downarrow})\right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right]\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\hat{c}^\dagger_{\mathbf{r}\sigma}\left[\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\right] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} - \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\right)\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\end{aligned} \tag{3.29}$$

We have a mirrored situation with respect to the Fock terms. For the Cooper channel, the s.s. EVs are zero if $\mathrm{U}^z(1)$ is unbroken, while the o.s. sector are not.

In Tab. 3.3 we summarise these considerations, listing a set of selection rules about the EVs of Eqns. (3.27), (3.23), dividing in the two sectors "o.s." and "s.s.". By identifying these selection rules, we set grounds for the detailed derivation of each chapter and heavily simplify application of HF model to the EHM. We now move to the last part of this chapter: reciprocal space description of the EHM.

3.3 The EHM in reciprocal space

The EHM is symmetric under discrete lattice translations. We will investigate phases that do not (completely) break translational invariance. Therefore, It is convenient to move to reciprocal space, by Fourier-transforming the hamiltonian. We will use the following convention:

$$\hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}}\sum_{\mathbf{k}\in\mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{r}}\hat{c}_{\mathbf{k}\sigma} \tag{3.30}$$

Throughout the text, $\overset{\mathcal{F}}{=}$ will indicate a passage where a Fourier transform is used. "BZ" is shorthand notation for the Brillouin Zone.

It is useful to recall that the following equation for Kronecker's delta holds:

$$\frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} e^{i\mathbf{k}\cdot\mathbf{r}} = \delta_{\mathbf{k}=\mathbf{0}} \tag{3.31}$$

On one sublattice:

$$\frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}/2} e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}=\mathbf{0}} \tag{3.32}$$

The notation $\mathcal{L}/2$ is shorthand to indicate $\mathcal{S}_1$ or $\mathcal{S}_2$.

Kinetic part. The kinetic operators transforms as follows:

$$\hat{H}_t \overset{\mathcal{F}}{=} \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma}\epsilon_{\mathbf{k}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{3.33}$$

where we defined the bare bands:

$$\epsilon_{\mathbf{k}} \equiv -2t\left(\cos k_x + \cos k_y\right) \tag{3.34}$$

Let us prove this first transform.

Proof 3.7 ($\mathcal{F}$-transform of $\hat{H}_t$).

Consider the kinetic operator $\hat{H}_t$:

$$\hat{H}_t = -t\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}\right]$$

Applying the convention of Eq. (3.30), we get:

$$\begin{aligned}
\hat{H}_t &\overset{\mathcal{F}}{=} -t\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left(\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}\in\mathrm{BZ}}e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\right)\left(\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}'\in\mathrm{BZ}}e^{-i\mathbf{k}'\cdot\mathbf{r}'}\hat{c}_{\mathbf{k}'\sigma}\right) + \text{h.c.}\\
&= -2t\sum_{\mathbf{k},\mathbf{k}'}\sum_{\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\times\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{L}/2}e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}\times\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}e^{i\mathbf{k}'\cdot\boldsymbol{\delta}}
\end{aligned}$$

We used Eq. (3.24) and recognised that the h.c. term gives an identical contribution. We recognise in the central term the relation of Eq. (3.32). Hence,

$$\begin{aligned}
\hat{H}_t &\overset{\mathcal{F}}{=} -2t\sum_{\mathbf{k},\mathbf{k}'}\sum_{\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\left[\frac{e^{ik_x}+e^{-ik_x}}{2}+\frac{e^{ik_y}+e^{-ik_y}}{2}\right]\\
&= \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma}\epsilon_{\mathbf{k}}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}
\end{aligned}$$

which proves Eq. (3.33).

Local repulsion. The local repulsion transforms as follows:

$$\hat{H}_U \overset{\mathcal{F}}{=} \frac{U}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{3.35}$$

where sums over the variables $\mathbf{K}, \mathbf{k}, \mathbf{k}'$ must be intended as over the BZ.

Proof 3.8 ($\mathcal{F}$-transform of $\hat{H}_U$).

Using the convention of Eq. (3.30), we get for $\hat{H}_U$:

$$\begin{aligned}
\hat{H}_U &= U\sum_{\mathbf{r}\in\mathcal{L}}\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\\
&\overset{\mathcal{F}}{=} U\sum_{\mathbf{r}\in\mathcal{L}}\frac{1}{(L_xL_y)^2}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}e^{i\mathbf{k}_1\cdot\mathbf{r}}\hat{c}^{\dagger}_{\mathbf{k}_1\uparrow}e^{i\mathbf{k}_2\cdot\mathbf{r}}\hat{c}^{\dagger}_{\mathbf{k}_2\downarrow}e^{-i\mathbf{k}_3\cdot\mathbf{r}}\hat{c}_{\mathbf{k}_3\downarrow}e^{-i\mathbf{k}_4\cdot\mathbf{r}}\hat{c}_{\mathbf{k}_4\uparrow}\\
&= \frac{U}{L_xL_y}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}\hat{c}^{\dagger}_{\mathbf{k}_1\uparrow}\hat{c}^{\dagger}_{\mathbf{k}_2\downarrow}\hat{c}_{\mathbf{k}_3\downarrow}\hat{c}_{\mathbf{k}_4\uparrow}\times\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{L}}e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}}\\
&= \frac{U}{L_xL_y}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}\hat{c}^{\dagger}_{\mathbf{k}_1\uparrow}\hat{c}^{\dagger}_{\mathbf{k}_2\downarrow}\hat{c}_{\mathbf{k}_3\downarrow}\hat{c}_{\mathbf{k}_4\uparrow}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}
\end{aligned}$$

where we used Eq. (3.31). Let $\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4=\mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$\mathbf{k}_1 \equiv \mathbf{K}+\mathbf{k} \qquad \mathbf{k}_2 \equiv \mathbf{K}-\mathbf{k} \qquad \mathbf{k}_3 \equiv \mathbf{K}-\mathbf{k}' \qquad \mathbf{k}_4 \equiv \mathbf{K}+\mathbf{k}'$$

Substituting, we get the expression of Eq. (3.35).

Non-local attraction. In o.s. sector, the non-local attraction transforms as:

$$\hat{H}_V^{(\text{o.s.})} \overset{\mathcal{F}}{=} -\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{3.36}$$

In the s.s. sector:

$$\begin{aligned} \hat{H}_V^{(\text{s.s.})} \overset{\mathcal{F}}{=} -\frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} & \left[\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right] \\ & \times \left[\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\uparrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\downarrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow}\right] \end{aligned} \tag{3.37}$$

Sums over the variables $\mathbf{K}, \mathbf{k}, \mathbf{k}'$ must be intended as over the BZ.

Proof 3.9 ($\mathcal{F}$-transform of $\hat{H}_V$).

To prove Eqns. (3.36), (3.37), it is useful to start by defining:

$$h_{\mathbf{r}\sigma\sigma'} \equiv -V \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right)$$

with:

$$\hat{H}_V = \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'}$$

The product of densities on neighbouring sites transforms as:

$$\begin{aligned} \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\sigma} \\ &\overset{\mathcal{F}}{=} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \end{aligned}$$

Then:

$$\begin{aligned} h_{\mathbf{r}\sigma\sigma'} &\overset{\mathcal{F}}{=} -\frac{V}{(L_x L_y)^2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\ &\qquad\qquad \times \left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} + e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\right) \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \\ &= -\frac{2V}{(L_x L_y)^2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}\right] \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{L}/2$ (which is, one sublattice). Applying Eq. (3.32) gives back momentum conservation

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}$$

where we used Eq. (3.32). Define $\mathbf{k}$, $\mathbf{k}'$ and $\mathbf{K}$ as we did fo the local repulsion, and

$$\delta\mathbf{k} \equiv \mathbf{k} - \mathbf{k}'$$

Hence:

$$
\begin{aligned}
\hat{H}_V &= \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'} \\
&\overset{\mathcal{F}}{=} -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \cos\left(\delta\mathbf{k}\cdot\boldsymbol{\delta}\right)\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\
&= -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}
\end{aligned}
\tag{3.38}
$$

Finally, we can separate in same spin and opposite spin channels. This gives:

$$
\begin{aligned}
\hat{H}_V \overset{\mathcal{F}}{=} &\overbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}} \\
&\underbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}
\end{aligned}
$$

The s.s. part confirms Eq. (3.37). The o.s. part can be simplified: since

$$
\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}
$$

due to fermionic anti-commutation rules, and since the composite transform

$$
\mathbf{k}\to-\mathbf{k} \qquad \mathbf{k}'\to-\mathbf{k}'
$$

leaves the form factor $\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)$ invariant, we reduce the spin sum to two times the $\sigma=\uparrow$ part. This confirms Eq. (3.36)

In reciprocal space, following the general decomposition presented in Eq. (**??**) for a quartic operator $\hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4}$, we distinguish contractions as follows:

- Hartree contractions:

$$
\hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4} \qquad \hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4}
$$

- Fock contractions:

$$
\hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4} \qquad \hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4}
$$

- Cooper contractions:

$$
\hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4} \qquad \hat{c}^\dagger_{\mathbf{k}_1}\hat{c}^\dagger_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4}
$$

with appropriate sign factors.

3.4 Square lattice spatial symmetry channels

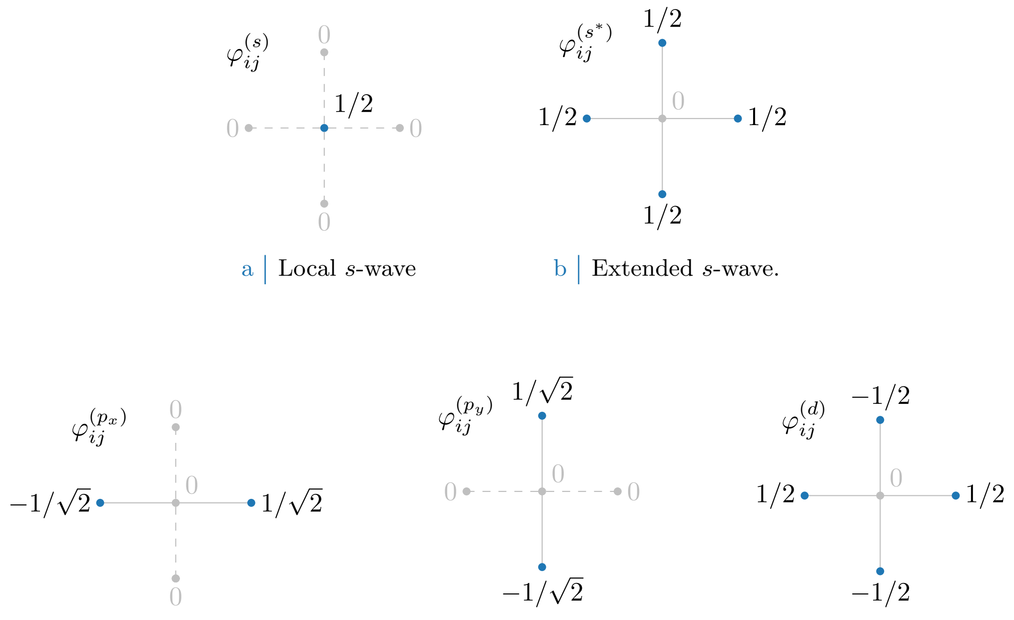
In this section we give two mathematical identities that are going to be used extensively. We define the set of functions $\varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'}$, with $\gamma\in\{s,s^*,p_x,p_y,d\}$, in Tab. 3.4; we plot them in Fig. 3.4. We will often refer to them as "harmonics". By construction, these functions obey an orthonormality condition:

$$
\frac{1}{L_xL_y}\sum_{\mathbf{r}\mathbf{r}'}\varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'}\varphi^{(\gamma')*}_{\mathbf{r}\mathbf{r}'} = \delta_{\gamma\gamma'} \tag{3.39}
$$

which is proven by direct computation.

Channel	Form factor	Graph
s-wave	$\varphi^{(s)}_{\mathbf{rr}'} = \delta_{\mathbf{rr}'}$	Fig. 3.4a
s^*-wave	$\varphi^{(s^*)}_{\mathbf{rr}'} = \frac{1}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 3.4b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{rr}'} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}}\right]$	Fig. 3.4c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{rr}'} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 3.4d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{rr}'} = \frac{1}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 3.4e

Table 3.4 | The harmonics $\varphi^{(\gamma)}_{\mathbf{rr}'}$. The notation of Fig. 3.3 is used. In the last column, we indicate a graphic representation given in the subfigures Fig. 3.4.



a | Local s-wave

b | Extended s-wave.

c | p_x-wave.

d | p_y-wave.

e | $d_{x^2-y^2}$-wave.

Figure 3.4 | The harmonics listed in Tab. 3.4. In figures five sites are represented: the hub site i and its four NNs. Solid lines connect the hub sites to a rim site where the harmonic is non-zero. Dashed lines connect to a rim site where the harmonic is zero.

Decomposition in symmetry channels. Consider a generic function of two variables $f(\mathbf{r}, \mathbf{r}')$. Suppose that, for a given centre $\mathbf{r}$, the function is non-zero only on the site itself ($\mathbf{r}' = \mathbf{r}$) and its NNs $\mathbf{r}' = \mathbf{r} \pm \boldsymbol{\delta}$, with $\boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$. In other words,

$$f(\mathbf{r}, \mathbf{r}') \neq 0 \quad \iff \quad |\mathbf{r} - \mathbf{r}'| = 0, 1 \tag{3.40}$$

in units of the lattice step. Using the harmonics $\varphi^{(\gamma)}_{\mathbf{rr}'}$, we get the following decomposition:

$$f(\mathbf{r}, \mathbf{r}') = \sum_{\gamma} f^{(\gamma)} \varphi^{(\gamma)}_{\mathbf{rr}'} \tag{3.41}$$

where we defined:

$$f^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{r},\mathbf{r}'} f(\mathbf{r}, \mathbf{r}') \varphi^{(\gamma)}_{\mathbf{rr}'} \tag{3.42}$$

Eq. (3.41) will be referred to as "decomposition in symmetry channels". The sum over γ is extended to the five symmetries of Tab. 3.4.

Proof 3.10 (Decomposition in symmetry channels of Eq. (3.41)).

Consider a function $f(\mathbf{r}, \mathbf{r}')$ defined as in Eq. (3.40) and the harmonics of Tab. 3.4. By direct computation, the following identity holds

$$\begin{aligned} f(\mathbf{r}, \mathbf{r}+\underline{\mathbf{x}}) &= \sum_{\mathbf{r}' \in \mathcal{L}} f(\mathbf{r}, \mathbf{r}') \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} \\ &= \sum_{\mathbf{r}' \in \mathcal{L}} f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} + \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} + \sqrt{2} \varphi^{(p_x)}_{\mathbf{r}\mathbf{r}'} \right) \end{aligned}$$

Similarly, also the following relations hold

$$\begin{aligned} f(\mathbf{r}, \mathbf{r}) &= \sum_{\mathbf{r}' \in \mathcal{L}} f(\mathbf{r}, \mathbf{r}') \varphi^{(s)}_{\mathbf{r}\mathbf{r}'} \\ f(\mathbf{r}, \mathbf{r}-\underline{\mathbf{x}}) &= \sum_{\mathbf{r}' \in \mathcal{L}} f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} + \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} - \sqrt{2} \varphi^{(p_x)}_{\mathbf{r}\mathbf{r}'} \right) \\ f(\mathbf{r}, \mathbf{r}+\underline{\mathbf{y}}) &= \sum_{\mathbf{r}' \in \mathcal{L}} f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} - \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} + \sqrt{2} \varphi^{(p_y)}_{\mathbf{r}\mathbf{r}'} \right) \\ f(\mathbf{r}, \mathbf{r}-\underline{\mathbf{y}}) &= \sum_{\mathbf{r}' \in \mathcal{L}} f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} - \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} - \sqrt{2} \varphi^{(p_y)}_{\mathbf{r}\mathbf{r}'} \right) \end{aligned}$$

Hence, by defining

$$\begin{aligned} f^{(s)} &\equiv f(\mathbf{r}, \mathbf{r}) \\ f^{(s^*)} &\equiv f(\mathbf{r}, \mathbf{r}+\underline{\mathbf{x}}) + f(\mathbf{r}, \mathbf{r}-\underline{\mathbf{x}}) + f(\mathbf{r}, \mathbf{r}+\underline{\mathbf{y}}) + f(\mathbf{r}, \mathbf{r}-\underline{\mathbf{y}}) \\ f^{(p_x)} &\equiv f(\mathbf{r}, \mathbf{r}+\underline{\mathbf{x}}) - f(\mathbf{r}, \mathbf{r}-\underline{\mathbf{x}}) \\ f^{(p_y)} &\equiv f(\mathbf{r}, \mathbf{r}+\underline{\mathbf{y}}) - f(\mathbf{r}, \mathbf{r}-\underline{\mathbf{y}}) \\ f^{(d)} &\equiv f(\mathbf{r}, \mathbf{r}+\underline{\mathbf{x}}) + f(\mathbf{r}, \mathbf{r}-\underline{\mathbf{x}}) - f(\mathbf{r}, \mathbf{r}+\underline{\mathbf{y}}) - f(\mathbf{r}, \mathbf{r}-\underline{\mathbf{y}}) \end{aligned}$$

we get the decomposition:

$$f(\mathbf{r}, \mathbf{r}') = \sum_{\gamma} f^{(\gamma)} \varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'}$$

The function f depends only on the difference of its two spatial variables. In other words, all the $f^{(\gamma)}$ components we defined above are independent of $\mathbf{r}$. Hence we can redefine them as in Eq. (3.42), averaging over $\mathbf{r}$. The decomposition obtained above still holds. With this redefinition of the $f^{(\gamma)}$ components, we recognise Eq. (3.41).

We decomposed the original function in its harmonics. Moving to reciprocal space, we get the $\mathcal{F}$-transformed functions of Tab. 3.5, also satisfying the orthonormality condition:

$$\frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma')*}_{\mathbf{k}} = \delta_{\gamma\gamma'}$$

The functions of Tab. 3.5 are plotted in the BZ in Fig. 3.5.

A useful identity. We conclude this chapter by proving a useful mathematical identity:

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2} \sum_{\gamma \neq s} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} \tag{3.43}$$

$$= \frac{1}{2} \sum_{\gamma \neq s} \varphi^{(\gamma)*}_{\mathbf{k}} \varphi^{(\gamma)}_{\mathbf{k}'} \tag{3.44}$$

Structure	Structure factor	Graph
s-wave	$\varphi^{(s)}_{\mathbf{k}} = 1$	Fig. 3.4a
Extended s-wave	$\varphi^{(s^*)}_{\mathbf{k}} = \cos k_x + \cos k_y$	Fig. 3.4b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{k}} = i\sqrt{2}\sin k_x$	Fig. 3.4c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{k}} = i\sqrt{2}\sin k_y$	Fig. 3.4d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{k}} = \cos k_x - \cos k_y$	Fig. 3.4e

Table 3.5 | $\mathcal{F}$-transforms of the harmonics of Tab. 3.4. These functions are plotted in Fig. 3.5.

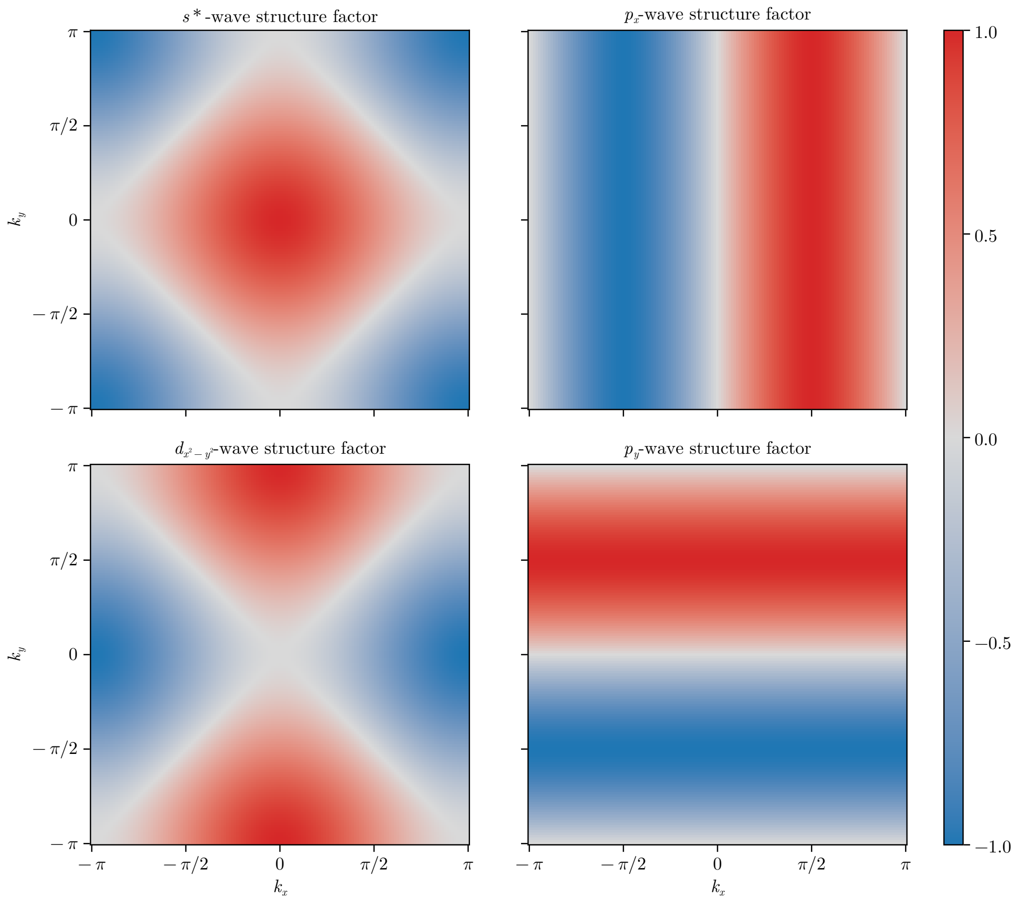


Figure 3.5 | Representation of the various structure factors listed in Tab. 3.5 on the BZ. For the s^* and $d_{x^2-y^2}$-wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted.

Note that in both rows above the sum is intended for $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$, excluding s.

Proof 3.11 (Proof of Eq. (3.43)).

Let $k_\ell \in \mathbb{R}$, with $\ell \in \{x, y\}$. We define:

$$c_\ell \equiv \cos k_\ell \qquad s_\ell \equiv \sin k_\ell \qquad c'_\ell \equiv \cos k'_\ell \qquad s'_\ell \equiv \sin k'_\ell$$

Let $\delta k_\ell \equiv k_\ell - k'_\ell$. Using basic trigonometric identities we get:

$$\cos(\delta k_x) + \cos(\delta k_y) = \underbrace{(c_x c'_x + c_y c'_y)}_{\text{Symmetric}} + \underbrace{(s_x s'_x + s_y s'_y)}_{\text{Anti-symmetric}} \tag{3.45}$$

All four terms are invariant under simultaneous inversion of both momenta, $(\mathbf{k}, \mathbf{k}') \to (-\mathbf{k}, -\mathbf{k}')$. Instead, if we invert just one momentum of the pair, the first two are symmetric for both arguments $\mathbf{k}$, $\mathbf{k}'$; the second two are anti-symmetric. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$\begin{aligned}
\cos(\delta k_x) + \cos(\delta k_y) = {} & \frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^*\text{-wave}) \\
& + s_x s'_x && (p_x\text{-wave}) \\
& + s_y s'_y && (p_y\text{-wave}) \\
& + \frac{1}{2}(c_x - c_y)(c'_x - c'_y) && (d_{x^2-y^2}\text{-wave})
\end{aligned}$$

which is written in compact form as in Eq. (3.43).

Chapter 4

The normal phase

In this chapter we apply the HF methods sketched in Sec. 2.2 to the "normal phase". We are studying the model in a state were no symmetry of those enumerated in Tab. 3.1 is broken. The application of HF methods in this context will lead to a self-consistent renormalisation of the hopping parameter t, which will produce a Mott localisation effect.

This chapter is divided in three parts. First, we discuss the symmetries of the normal phase, following the general discussion of Chap. 3. Second, we apply HF methods to derive the self-consistent approximated hamiltonian and impose some theoretical constraints on the solution. Lastly, we present numeric results.

4.1 The normal phase and its symmetries

In the normal phase no symmetry is broken. Thus, the HF wavefunction approximating the ground state must exhibit the full $\mathrm{U}(2) \otimes \mathrm{C}_4 \otimes \mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$ symmetry discussed in Sec. 3.1. Translational invariance and spin rotational symmetry imply that the site occupation is independent of site and spin indices,

$$\langle \hat{n}_{i\sigma} \rangle = n \tag{4.1}$$

with $n \in [0, 1]$.

We define the single-state occupation operator in reciprocal space:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}$$

and its EV

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{n}_{\mathbf{k}\sigma} \rangle \qquad u_{\mathbf{k}\sigma} = u^*_{\mathbf{k}\sigma} \tag{4.2}$$

The relation on the right follows from the fact that $\hat{n}_{\mathbf{k}\sigma}$ is hermitian by construction. Note that $\hat{n}_{\mathbf{k}\sigma}$ is not the Fourier-transform of $\hat{n}_{i\sigma}$. Translational invariance implies the following result:

$$\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma'} \rangle = \delta_{\mathbf{k}=\mathbf{k}'} \delta_{\sigma=\sigma'} \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle \tag{4.3}$$

Proof 4.1 (Proof of Eq. (4.3)).

Consider the generic EV

$$\left\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma'} \right\rangle$$

for two momenta $\mathbf{k}$, $\mathbf{k}'$. Moving to real space via an $\mathcal{F}$-transform, we get

$$\left\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma'} \right\rangle \overset{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} \sum_{\mathbf{r}'\in\mathcal{L}} e^{-i\left(\mathbf{k}\cdot\mathbf{r} - \mathbf{k}'\cdot\mathbf{r}'\right)} \left\langle \hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}^\dagger_{\mathbf{r}'\sigma'} \right\rangle$$

where we used the convention of Eq. (3.30). Let now $\Delta\mathbf{r} \equiv \mathbf{r}_2 - \mathbf{r}_1$. Translational invariance implies that

$$g(\Delta\mathbf{r}) = \left\langle \hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}^\dagger_{\mathbf{r}+\Delta\mathbf{r}\sigma'} \right\rangle$$

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalisation and resymmetrization

Table 4.1 | Relevant Wick's contractions and net effect in the normal phase.

is independent of $\mathbf{r}$. Thus we have:

$$\begin{aligned}\langle \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}'\sigma'}\rangle &\overset{\mathcal{F}}{=} \sum_{\Delta\mathbf{r}\in\mathcal{L}} e^{-i\mathbf{k}'\cdot\Delta\mathbf{r}} g(\Delta\mathbf{r}) \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{-i\left(\mathbf{k}-\mathbf{k}'\right)\cdot\mathbf{r}} \\ &= \sum_{\Delta\mathbf{r}\in\mathcal{L}} e^{-i\mathbf{k}'\cdot\Delta\mathbf{r}} g(\Delta\mathbf{r})\delta_{\mathbf{k}=\mathbf{k}'}\end{aligned}$$

where we used the identity of Eq. (3.31). Then the above EV vanishes for $\mathbf{k} \neq \mathbf{k}'$. This proves Eq. (4.3).

We anticipate here that we will actually allow for $C_4 \overset{\text{SSB}}{\to} C_2$, namely, breaking of planar rotational symmetry. As will result from our discussion, the normal phase does not break C_4 even if we allow it to. Nevertheless we carry out this general derivation because breaking of planar rotational symmetry will be observed in Chap. **??** in the context of superconductivity, and the computations carried out for the normal phase will be re-used.

Applying the HF methods to the interactions $\hat{H}_U$ and $\hat{H}_V$, from Tabs. 3.2 and 3.3, we know that in the normal phase the allowed contractions are:

$\hat{H}_U$	Hartree contractions
$\hat{H}_V$ (o.s. sector)	Hartree contractions
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

As we will see in detail and is summarised in Tab. 4.1, Hartree contractions give a chemical potential shift, while the Fock contraction renormalises the bare bands.

In the normal phase, spin is a pure physical degeneracy, being the $\uparrow$ and $\downarrow$ sectors separated; moreover, $\hat{H}$ is $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. This implies that all physical quantities, as is for the normal phase, should not depend on the spin index. We will perform all computations leaving σ explicit and using this consideration at the end.

4.2 Application of HF methods to the normal phase

The quartic part of $\hat{H}$ is given by the the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. Applying HF methods, we get the decomposition of Eqns. (2.20), (2.21). In the next section we analyse the Hartree channel, which contains all Hartree-like terms. We will see that $\hat{H}_U$ and $\hat{H}_V$ are approximated, in this channel, by a multiple of the number operator. In the section that follows, we deal with Fock terms, and show how $\hat{H}_V$ is approximated, in that channel, by an operator that is formally equivalent to the kinetic operator $\hat{H}_t$.

4.2.1 Hartree effects: chemical potential renormalisation

Consider the Hartree channel of Eqns. (2.20), (2.21),

$$\hat{H}_U \overset{\text{HF}}{\simeq} U\sum_i \left(\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\langle\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\rangle + \langle\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow} - \langle\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle\langle\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\rangle\right) + (\text{all the rest})$$

$$\hat{H}_V \overset{\text{HF}}{\simeq} -V\sum_{\langle ij\rangle}\sum_{\sigma,\sigma'} \left(\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\langle\hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\rangle + \langle\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\rangle\hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'} - \langle\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\rangle\right) + (\text{all the rest})$$

Let us deal with those separately.

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ decomposes in Hartree channel as specified in Eq. (2.20)

$$\hat{H}_U \overset{\text{HF}}{\simeq} U \sum_i \left(\langle \hat{n}_{i\uparrow} \rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow} \langle \hat{n}_{i\downarrow} \rangle - \langle \hat{n}_{i\downarrow} \rangle \langle \hat{n}_{i\uparrow} \rangle \right)$$

For a translationally invariant phase such as the normal phase, it holds

$$\langle \hat{n}_{i\sigma} \rangle = n \tag{4.4}$$

which gives

$$\begin{aligned} \hat{H}_U &\overset{\text{HF}}{\simeq} nU \sum_i \left(\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow} \right) - U \sum_i n^2 \\ &= nU \times \hat{N} - E^{(\text{N})}_{\text{H}/U} \end{aligned} \tag{4.5}$$

being $\hat{N}$ the total number operator and $E^{(\text{N})}_{\text{H}/U}$ the "local Hartree contraction energy" (also known as double counting term),

$$E^{(\text{N})}_{\text{H}/U} = L_x L_y \times n^2 U \tag{4.6}$$

This energy contributes negatively to the ground-state energy, for $U > 0$. The chemical potential is corrected by

$$\mu \to \mu - nU \tag{4.7}$$

The non-local attraction further corrects μ.

Non-local attraction. The non-local attraction is divided in two sectors, as in Eq. (3.21). From Eq. (2.21), its Hartree-like terms are:

$$\begin{aligned} \hat{H}_V \overset{\text{HF}}{\simeq} & \overbrace{-V \sum_{\langle ij \rangle} \sum_\sigma \left(\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma} \langle \hat{n}_{j\sigma} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle \right)}^{\text{s.s.}} \\ & \underbrace{-V \sum_{\langle ij \rangle} \sum_\sigma \left(\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma} \langle \hat{n}_{j\bar{\sigma}} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle \right)}_{\text{o.s.}} + (\text{all the rest}) \end{aligned}$$

where "all the rest" collects all non-Hartree terms. From Eq. (4.4), we get

$$\hat{H}_V \overset{\text{HF}}{\simeq} \underbrace{-nV \sum_{\langle ij \rangle} \sum_\sigma \left(\hat{n}_{j\sigma} + \hat{n}_{i\sigma} \right)}_{\text{s.s.}} \underbrace{-nV \sum_{\langle ij \rangle} \sum_\sigma \left(\hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma} \right)}_{\text{o.s.}} - E^{(\text{N})}_{\text{H}/V} + (\text{all the rest}) \tag{4.8}$$

with $E^{(\text{N})}_{\text{H}/V}$ the normal state shift to energy brought by $\hat{H}_V$:

$$\begin{aligned} E^{(\text{N})}_{\text{H}/V} &= -V \sum_{\langle ij \rangle} \sum_\sigma \left(\langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle + \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle \right) \\ &= -2V \sum_{\langle ij \rangle} \sum_\sigma n^2 \\ &= -L_x L_y \times 2zn^2 V \end{aligned} \tag{4.9}$$

Note the similarity with Eq. (4.6).

Getting back to Eq. (4.8), both sums give a number operator. Note that we sum over NNs: each site is connected to $z = 4$ sites. Including this combinatorial factor, we have

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E^{(\text{N})}_{\text{H}/V} + (\text{all the rest}) \tag{4.10}$$

This accounts for the final Hartree shift of μ,

$$\mu \to \mu + 2znV \tag{4.11}$$

Collecting everything, we get the HF approximation to the two interactions in the Hartree channel:

$$\hat{H}_U + \hat{H}_V \overset{\text{HF}}{\simeq} -n(2zV - U)\hat{N}$$

Considering the net shift to μ due to both interactions by combining Eqns. (4.7), (4.11), we get the final RMP anticipated in Tab. 4.1,

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{4.12}$$

This result will remain valid throughout all phases discussed in this text.

4.2.2 Fock effects: bare bands renormalisation

We move to Fock terms. From Wick's decomposition of $\hat{H}_V$, the unique allowed Fock term comes from the same-spin sector. This selection rules is implied by the SU(2) symmetry of the normal phase and was stated in Tab. 3.3. Then, from Eq. (2.21):

$$\hat{H}_V \overset{\text{HF}}{\simeq} (\text{Hartree channel}) + V \sum_{\langle ij \rangle} \sum_{\sigma} \left(\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right) \tag{4.13}$$

Note the plus sign in front of the displayed term: it comes from the minus sign in Fock term of Eq. (2.1).

Consider the Fourier Transform $\mathcal{F}$ given in Eq. (3.37), which gives

$$\begin{aligned} \hat{H}_V \overset{\text{HF}}{\simeq} & (\text{Hartree channel}) \\ & + \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left(\cos(\delta k_x) + \cos(\delta k_y)\right) \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E^{(\text{N})}_{\text{F}/V} \end{aligned} \tag{4.14}$$

Here, the 2 prefactor comes from the fact that the first two operators in square brackets in Eq. (4.13) are equal in reciprocal space. The double counting energy shift is given by

$$E^{(\text{N})}_{\text{F}/V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left(\cos(\delta k_x) + \cos(\delta k_y)\right) \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \tag{4.15}$$

Recall the definition of $u_{\mathbf{k}\sigma}$ given in Eq. (4.2). Using the functions of Tab. 3.5, we define its harmonics:

$$u^{(\gamma)}_\sigma \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle \tag{4.16}$$

The non-local attraction $\hat{H}_V$ is approximated, in Fock channel, by:

$$\hat{H}_V \overset{\text{HF}}{\simeq} (\text{Hartree channel}) + V \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \left(u^{(s^*)}_\sigma \varphi^{(s^*)*}_{\mathbf{k}} + u^{(d)}_\sigma \varphi^{(d)*}_{\mathbf{k}} \right) \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - E^{(\text{N})}_{\text{F}/V} \tag{4.17}$$

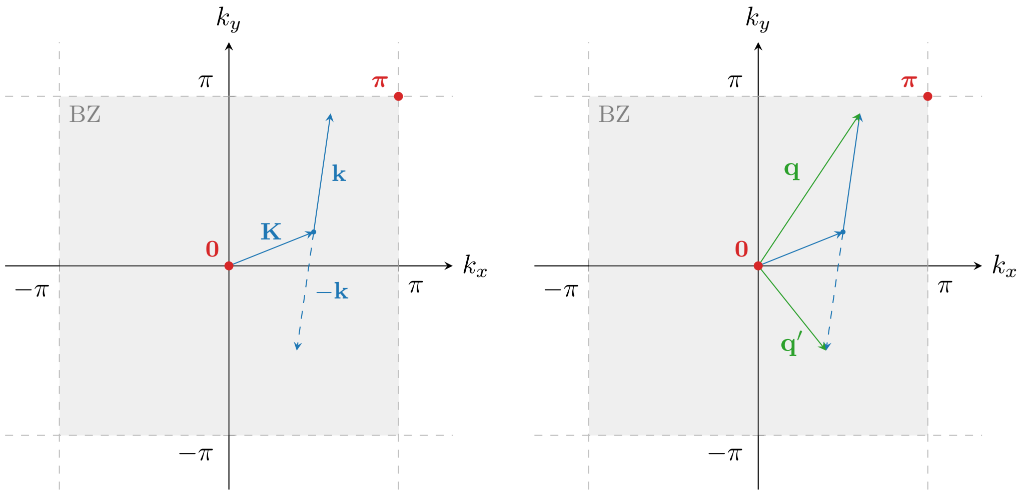
We provide a proof.

Proof 4.2 (Proof of Eq. (4.17)).

Consider Eq. (4.14). Applying Eq. (4.3), it becomes:

$$\begin{aligned} \hat{H}_V \overset{\text{HF}}{\simeq} & (\text{Hartree channel}) \\ & + \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left(\cos 2k_x + \cos 2k_y\right) \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} - E^{(\text{N})}_{\text{F}/V} \end{aligned} \tag{4.18}$$

Let now $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. A representation of this variables change is given in Fig. 4.1.



a | Generic vectors.

b | Change of variables.

Figure 4.1 | Representation of the variable change employed to pass from Eq. (4.18) to (4.19). In Fig. 4.1a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in$ BZ. In Fig. 4.1b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

Being (for $\ell = x, y$)

$$\begin{aligned} \delta q_\ell &\equiv q_\ell - q'_\ell \\ &= (K_\ell + k_\ell) - (K_\ell - k_\ell) \\ &= 2k_\ell \end{aligned}$$

we reduce Eq. (4.14) to

$$\begin{aligned} \hat{H}_V \overset{\text{HF}}{\simeq} &\text{ (Hartree channel)} \\ &+ \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma \left[\cos(\delta q_x) + \cos(\delta q_y)\right] \underbrace{\langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle}_{u_{\mathbf{q}\sigma}} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{4.19}$$

where we recognised $u_{\mathbf{q}\sigma}$, as was defined in Eq. (4.2). Using Eq. (3.43), we obtain

$$\begin{aligned} \hat{H}_V \overset{\text{HF}}{\simeq} &\text{ (Hartree channel)} \\ &+ \sum_{\gamma,\sigma} \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right) \times V \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \end{aligned} \tag{4.20}$$

where we used the functions defined in Tab. 3.5. We recognise the harmonics $u^{(\gamma)}_\sigma$ from Eq. (4.16),

$$\hat{H}_V \overset{\text{HF}}{\simeq} \text{(Hartree channel)} + V \sum_{\mathbf{q}'\in\mathrm{BZ}} \sum_{\gamma,\sigma} u^{(\gamma)}_\sigma \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{4.21}$$

which is almost Eq. (4.17).

To complete the proof, we recognise that in the normal phase inversion symmetry is not broken, thus

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma}$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \dfrac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. 3.4b
$d_{x^2-y^2}$-wave	$u^{(d)} = \dfrac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. 3.4e

Table 4.2 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

This follows from the fact that in the normal phase the EV of the operator $\hat{n}_{\mathbf{k}\sigma}$ must be inversion-symmetric. It follows that antisymmetric harmonics are zero,

$$u_\sigma^{(p_\ell)} = 0 \quad \text{for} \quad \ell \in \{x, y\}$$

Using this observation in Eq. (4.21), we obtain Eq. (4.17).

Composing Eq. (4.17) with Eqns. (4.5), (4.10) we obtain the quadratic HF approximation to the EHM, for the normal phase. The chemical potential $\tilde{\mu}$ is fixed by density and determined during computations. To find the ground-state, we must determine self-consistently the two HFPs $u_\sigma^{(s^*)}$, $u_\sigma^{(d)}$ through the self-consistency Eq. (4.16). The symmetry-resolved equations are reported in Tab. 4.2.

The operator of Eq. (4.17) has the same algebraic structure of the kinetic operator $\hat{H}_t$ in reciprocal space, given in Eq. (3.33). Hence, we can collapse the two into one effective kinetic operator. Let the renormalised bare bands be

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)} V(\cos k_x + \cos k_y) + u_\sigma^{(d)} V(\cos k_x - \cos k_y) \tag{4.22}$$

In Fock channel, we have:

$$\hat{H}_t + \hat{H}_V \overset{\text{HF}}{\simeq} (\text{Hartree channel}) + \underbrace{\sum_{\mathbf{k}\in\mathrm{BZ}} \sum_\sigma \tilde{\epsilon}_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}}_{\hat{H}_{\tilde{t}}} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{4.23}$$

where we defined the effective kinetic operator, $\hat{H}_{\tilde{t}}$.

The renormalised bands of Eq. (4.22) are a sum of s^* and d-wave components. Instead, bare bands as defined Eq. (3.34) are purely s^*-symmetric functions,

$$\epsilon_{\mathbf{k}\sigma} = -2t(\cos k_x + \cos k_y) = -4t\varphi_{\mathbf{k}}^{(s^*)} \tag{4.24}$$

We define:

$$\tilde{t}_\sigma^{(s^*)} \equiv t - \frac{u_\sigma^{(s^*)} V}{2} \qquad \tilde{t}_\sigma^{(d)} \equiv -\frac{u_\sigma^{(d)} V}{2} \tag{4.25}$$

Hence, Eq. (4.22) can be written as follows:

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv -4\tilde{t}_\sigma^{(s^*)} \varphi_{\mathbf{k}}^{(s^*)} - 4\tilde{t}_\sigma^{(d)} \varphi_{\mathbf{k}}^{(d)} \tag{4.26}$$

Note the formal similarity with Eq. (4.24). Bands renormalisation is actually a renormalisation of the hopping parameter t,

$$t \ (s^*\text{-wave}) \quad \rightarrow \quad \tilde{t}_\sigma^{(s^*)} \ (s^*\text{-wave}) \text{ and } \tilde{t}_\sigma^{(d)} \ (d\text{-wave})$$

Therefore, t is an RMP. In Sec. 4.3.1 we will show how all of this is translated in real space.

We conclude this section by proving that the Fock contraction energy is given by:

$$E^{(\mathrm{N})}_{\mathrm{F}/V} = L_x L_y \times \frac{V}{2} \sum_\sigma \left(\left|u_\sigma^{(s^*)}\right|^2 + \left|u_\sigma^{(d)}\right|^2 \right) \tag{4.27}$$

Later, using spin degeneracy, we will omit the sum on σ and the $1/2$ prefactor.

Proof 4.3 (Non-local Fock contraction energy.).

Recall Eq. (4.15). Using Eq. (4.3), it reduces to:

$$E_{\mathrm{F}/V}^{(\mathrm{N})} = \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos(\delta q_x) + \cos(\delta q_y)\right] \langle \hat{c}_{\mathbf{q}\sigma}^{\dagger} \hat{c}_{\mathbf{q}\sigma} \rangle \langle \hat{c}_{\mathbf{q}'\sigma}^{\dagger} \hat{c}_{\mathbf{q}'\sigma} \rangle$$

Inserting Eq. (4.2), we get:

$$\begin{aligned} E_{\mathrm{F}/V}^{(\mathrm{N})} &= L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right) \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in \mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}'\sigma}^{*} \right) \\ &= L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left| u_{\sigma}^{(\gamma)} \right|^2 \end{aligned}$$

which, using the fact that $u_{\sigma}^{(p_\ell)} = 0$, proves Eq. (4.27).

In next section we collect all we found so far and discuss the properties of the HF hamiltonian.

4.3 The HF hamiltonian and its properties

In the normal phase, the HF approximation to the EHM hamiltonian is given by the following operator:

$$\hat{H} - \mu \hat{N} \overset{\mathrm{HF}}{\simeq} \sum_{\mathbf{k}\in \mathrm{BZ}} \sum_{\sigma} \tilde{\epsilon}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} - \tilde{\mu} \hat{N} - \left(E_{\mathrm{H}/U}^{(\mathrm{N})} + E_{\mathrm{H}/V}^{(\mathrm{N})} + E_{\mathrm{F}/V}^{(\mathrm{N})} \right) \tag{4.28}$$

Apart from energy shifts, this hamiltonian describes a perfectly degenerate Fermi gas whose bare bands are $\tilde{\epsilon}_{\mathbf{k}\sigma}$ as defined in Eq. (4.22), and chemical potential $\tilde{\mu}$ as defined in Eq. (4.12). The latter is completely determined by the model parameters; the former must be determined self-consistently. The ground state of the approximated hamiltonian obeys Fermi-Dirac statistics,

$$\langle \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}'\sigma'} \rangle = \delta_{\mathbf{k}=\mathbf{k}'} \delta_{\sigma=\sigma'} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \tag{4.29}$$

being f the Fermi-Dirac distribution,

$$f(\epsilon; \beta, \mu) \equiv \frac{1}{e^{\beta(\epsilon-\mu)} + 1}$$

Eq. (4.29) is the explicit version of Eq. (4.3).

In next subsection we discuss in detail the renormalisation of the hopping parameter in real space, with a focus on the meaning of the d-wave component. In the subsection that follow, we derive the free energy and entropy densities for the ground-state

4.3.1 Nematicity: d-wave bands and anisotropic hopping

In Eq. (4.23) we defined $\hat{H}_{\tilde{t}}$ in reciprocal space. Using Eq. (4.13) we get

$$\begin{aligned} \hat{H}_{\tilde{t}} &= \hat{H}_t + V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^{\dagger} \hat{c}_{i\sigma} + \text{h.c.} \right] \\ &= - \sum_{i,j} \sum_{\sigma} \left[\tilde{t}_{ij\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + \text{h.c.} \right] \end{aligned}$$

Note that in second line we are not summing over NNs, but on two site indices $i, j \in \mathcal{L}$. We defined the effective hopping amplitude

$$\tilde{t}_{ij\sigma} \equiv \frac{1}{2} \left(t - V \langle \hat{c}_{j\sigma}^{\dagger} \hat{c}_{i\sigma} \rangle \right) \qquad \text{(for } i, j \text{ NNs)}$$

The 1/2 prefactor was included to avoid double counting.

Due to translational invariance, the renormalised hopping $\tilde{t}_{ij\sigma}$ depends only on the difference $i-j$. Therefore we can use the decomposition of Eq. (3.41),

$$\tilde{t}_{ij\sigma} = \sum_{\gamma} \tilde{t}^{(\gamma)}_{\sigma} \varphi^{(\gamma)}_{ij}$$

The harmonics, given by Eq. (3.42), are identified with $\tilde{t}^{(s^*)}_{\sigma}$, $\tilde{t}^{(d)}_{\sigma}$ already introduced in Eq. (4.25). From the same argument we know that $\tilde{t}^{(p_x)}_{\sigma} = \tilde{t}^{(p_y)}_{\sigma} = 0$. Then, the renormalised hopping is explicitly given by

$$\tilde{t}_{\mathbf{r}\mathbf{r}'\sigma} = \frac{\tilde{t}^{(s^*)}_{\sigma}}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right] \\ + \frac{\tilde{t}^{(d)}_{\sigma}}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$$

Let us define:

$$\tilde{t}^{(x)}_{\sigma} \equiv \frac{\tilde{t}^{(s^*)}_{\sigma} + \tilde{t}^{(d)}_{\sigma}}{2} \tag{4.30}$$

$$\tilde{t}^{(y)}_{\sigma} \equiv \frac{\tilde{t}^{(s^*)}_{\sigma} - \tilde{t}^{(d)}_{\sigma}}{2} \tag{4.31}$$

With this definition, we can rewrite $\hat{H}_{\tilde{t}}$ as:

$$\hat{H}_{\tilde{t}} = -\sum_{\mathbf{r}\in\mathcal{L}}\sum_{\sigma}\left[\tilde{t}^{(x)}_{\sigma}\left(\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\underline{\mathbf{x}}\sigma} + \hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}-\underline{\mathbf{x}}\sigma}\right) + \tilde{t}^{(y)}_{\sigma}\left(\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\underline{\mathbf{y}}\sigma} + \hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}-\underline{\mathbf{y}}\sigma}\right) + \text{h.c.}\right]$$

This is what we intend for "nematic effects": the breaking of planar rotational symmetry, with the consequence that the kinetic operators in x and y directions differ. Breaking of planar rotational symmetry and uniform density are compatible.

Proof 4.4 (A d-wave hopping is compatible with uniform density).

> Consider the on-site occupation operator $\hat{n}_{\mathbf{r}\sigma} = \hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}$. Applying a Fourier transform as defined in Eq. (3.30), we obtain
>
> $$\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} \stackrel{\mathcal{F}}{=} \frac{1}{L_x L_y}\sum_{\mathbf{k},\mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}$$
>
> We compute the expectation value of the above operator and apply Eq. (4.29),
>
> $$\begin{aligned}\langle\hat{n}_{\mathbf{r}\sigma}\rangle &= \frac{1}{L_x L_y}\sum_{\mathbf{k},\mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\rangle \\ &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\text{BZ}} f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\end{aligned}$$
>
> The right-hand side of last line is independent of $\mathbf{r}$, therefore $\langle\hat{n}_{\mathbf{r}\sigma}\rangle$ is.

Note that we never used the actual topology of the renormalised bands $\tilde{\epsilon}_{\mathbf{k}}$, but only the fact that the ground-state obeys the Fermi-Dirac distribution. A perfectly degenerate Fermi gas is always translational invariant and thus cannot exhibit non-uniform symmetry.

A schematics of anisotropic hopping is given in Fig. 4.2: for each bond linking two sites, the hopping parameter can be a positive or negative combination of the two components, as the link direction is vertical or horizontal.

From now on, we will omit the spin index σ from both the renormalised bands and the renormalised hopping,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}} \qquad \forall\alpha \quad : \quad \tilde{t}^{(\alpha)}_{\sigma} \to \tilde{t}^{(\alpha)}$$

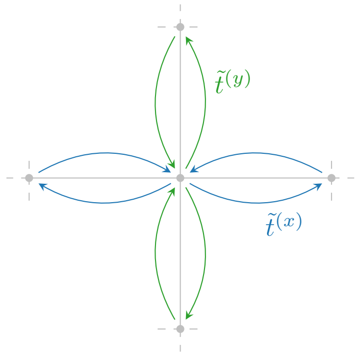


Figure 4.2 Graphic rendition of anisotropic hopping, with a s^*-wave component (symmetric by $\pi/$ rotations) and a d-wave component (antisymmetric).

The normal phase does not break spin rotational invariance. Therefore the renormalised bands must be independent of the spin index. Inverting Eq. (4.25), we can express $\tilde{t}^{(x)}$ and $\tilde{t}^{(y)}$ in terms of the HFPs $u^{(s^*)}$ and $u^{(d)}$

$$\tilde{t}^{(x)} \equiv t - \frac{u^{(s^*)} + u^{(d)}}{2} V$$
$$\tilde{t}^{(y)} \equiv t - \frac{u^{(s^*)} - u^{(d)}}{2} V$$

Inverting these relations we obtain:

$$u^{(s^*)} = \frac{\left(t - \tilde{t}^{(x)}\right) + \left(t - \tilde{t}^{(y)}\right)}{V} \tag{4.32}$$
$$u^{(d)} = \frac{\tilde{t}^{(x)} - \tilde{t}^{(y)}}{V} \tag{4.33}$$

It follows that if $u^{(d)} > 0$, the hopping in direction x is larger than in direction y is damped. If $u^{(d)} < 0$, roles are inverted.

4.3.2 The entropy density of the SC ground-state

In this section we compute the density of entropy of the ground state in the normal phase. Entropy density is given by

$$s^{(\mathrm{N})} = \frac{S^{(\mathrm{N})}}{L_x L_y}$$

being $S^{(\mathrm{N})}$ the total entropy.

The HF hamiltonian of Eq. (4.28) describes a non-interacting Fermi gas. The entropy density of a free Fermi gas in its ground state [29] is

$$s^{(\mathrm{N})} = -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \left[(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) \ln (1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \tag{4.34}$$

The 2 prefactor was included to account for spin degeneracy. The above equation can be written in another way:

$$s^{(\mathrm{N})} = -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \left[\ln (1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) - \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \tag{4.35}$$

Proof 4.5 (Proof of Eq. (4.35)).

The following identity

$$\ln\left(1-\frac{1}{e^x+1}\right) = \ln\left(\frac{e^x}{e^x+1}\right)$$
$$= x + \ln\left(\frac{1}{e^x+1}\right) \tag{4.36}$$

Substituting $x = \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})$and using Eq. (4.36) in Eq. (4.34) we obtain

$$s^{(\mathrm{N})} = -\frac{2k_B}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\Big[\ln\left(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$
$$-\beta(\tilde{\epsilon}_{\mathbf{k}}-\tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\Big]$$

Second and fourth terms cancel out. The remaining parts give Eq. (4.35).

4.3.3 The free energy density of the SC ground-state

In this section we compute the free energy density of the ground state, for the normal phase. The density of free energy is given by

$$f^{(\mathrm{N})} \equiv \frac{F^{(\mathrm{N})}}{L_xL_y} \tag{4.37}$$

being $F^{(\mathrm{N})}$ the total free energy,

$$F^{(\mathrm{N})} = E^{(\mathrm{N})}_{\mathrm{HF}} - \frac{S^{(\mathrm{N})}}{k_{\mathrm{B}}\beta} \tag{4.38}$$

Here $E^{(\mathrm{N})}_{\mathrm{HF}}$ is the ground-state energy for the HF hamiltonian. In the normal phase, we compute $f^{(\mathrm{N})}$ using the following formula:

$$f^{(\mathrm{N})} = \frac{2}{\beta L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\ln\left(1-f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \tilde{\mu}n \tag{4.39}$$

Proof 4.6 (Free energy density for the normal phase).

Using Eqns. (4.37), (4.38), we have

$$f^{(\mathrm{N})} = \frac{E^{(\mathrm{N})}_{\mathrm{HF}}}{L_xL_y} - \frac{s^{(\mathrm{N})}}{k_{\mathrm{B}}\beta}$$

with $s^{(\mathrm{N})}$ the entropy density computed in Eq. (4.35). Then, $f^{(\mathrm{N})}$ is obtained by summing the following terms:

1. The free particles ground state energy density,

$$e^{(\mathrm{N})}_{\mathrm{g}} = \frac{2}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\tilde{\epsilon}_{\mathbf{k}}-\tilde{\mu})f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

The prefactor 2 accounts for spin degeneracy.

2. The contraction energy density,

$$e^{(\mathrm{N})}_{\mathrm{c}} \equiv \frac{E^{(\mathrm{N})}_{\mathrm{c}}}{L_xL_y}$$
$$= -n^2U + V\left[2zn^2 - \left|u^{(s^*)}\right|^2 - \left|u^{(d)}\right|^2\right]$$

where in first line we included the all contraction energies from Eqns. (4.6), (4.9) and (4.27) in $E_{\mathrm{c}}^{(\mathrm{N})}$,

$$E_{\mathrm{c}}^{(\mathrm{N})} \equiv -\left[E_{\mathrm{H}/U}^{(\mathrm{N})} + E_{\mathrm{H}/V}^{(\mathrm{N})} + E_{\mathrm{F}/V}^{(\mathrm{N})}\right]$$

3. The entropic contribution to free energy

$$\begin{aligned} e_s^{(\mathrm{N})} &\equiv -\frac{s^{(\mathrm{N})}}{k_{\mathrm{B}}\beta} \\ &= \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right) - \frac{2}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \end{aligned}$$

   We used Eq. (4.35).

4. The chemical potential correction

$$e_n^{(\mathrm{N})} = \mu n$$

   We explained at the end of Sec. 2.3.1 why this correction is needed. Note here that we are using the "physical" value of μ, not its RMP counterpart $\tilde{\mu}$.

Thus, composing everything

$$\begin{aligned} f^{(\mathrm{N})} &\equiv e_{\mathrm{g}}^{(\mathrm{N})} + e_{\mathrm{c}}^{(\mathrm{N})} + e_s^{(\mathrm{N})} + e_n^{(\mathrm{N})} \\ &= \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \left[\mu + n(2zV - U)\right] n \end{aligned}$$

Using Eq. (4.12), we recognise

$$\mu + n(2zV - U) = \tilde{\mu}$$

Hence Eq. (4.39) is proven.

4.4 Analysis of the self-consistency equations

RESTART FROM HERE...

In this section we collect some results that follow from a formal analysis of the self-consistency problem for the normal phase. This phase is described by the HFPs $u^{(s^*)}$ and $u^{(d)}$. It is possible to prove that:

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \tag{4.40}$$

for any choice of $U, V > 0$ and doping δ. The proof follows from the detailed analysis of the self-consistency equations of Tab. 4.2, and is reported in App. A. Eq. (A.12) implies that in the normal phase there is no nematic effect, and the renormalised hopping parameter $\tilde{t}_{ij}$ is isotropic with a smaller amplitude than t,

$$0 < \tilde{t}^{(s^*)} < t$$

Here we used Eq. (4.25). Then, bands renormalisation

$$\epsilon_{\mathbf{k}} \quad \to \quad \tilde{\epsilon}_{\mathbf{k}}$$

is actually a flattening. We denote this effect as "shrinking".

Let us define the Magnetic Brillouin Zone (MBZ):

$$\mathrm{MBZ} \equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0\right\} \tag{4.41}$$

The MBZ coincides with the set of points (k_x, k_y) that satisfy the relation

$$|k_x| + |k_y| \leq \pi$$

We draw the MBZ in Fig. 4.3. The renormalised bands satisfy

$$\tilde{\epsilon}_{\mathbf{k}} = -4\tilde{t}^{(s^*)}\varphi_{\mathbf{k}}^{(s^*)}$$

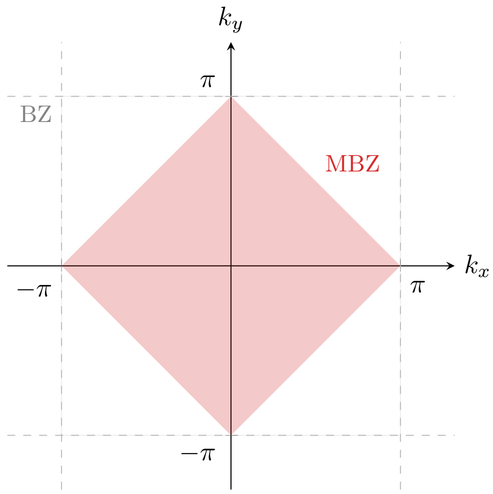


Figure 4.3 | Representation of the MBZ. The dashed lines represent the boundaries of the BZ.

Hence, $\tilde{\epsilon}_{\mathbf{k}} < 0$ inside the MBZ and $\tilde{\epsilon}_{\mathbf{k}} > 0$ outside. Along the boundary ∂MBZ we have $\tilde{\epsilon}_{\mathbf{k}} = 0$.

We conclude this section by proving some analytic properties of the unique HFP of the normal phase, $u^{(s^*)}$. Its self-consistency equation, from Tab. 4.2, reads

$$u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \tag{4.42}$$

We wrote $u^{(s^*)}$ as a function of chemical potential and temperature. Note that the renormalised bands, inside the Fermi-Dirac factor on the right-hand side, depend on $u^{(s^*)}$. We now prove that in the low-temperature limit the above equation implies that $u^{(s^*)}$ is a monotonically descendent function of doping.

Proof 4.7 ($u^{(s^*)}$ decreases with doping at low temperature).

We work at low temperature, $\beta \to 0$. The Fermi-Dirac distribution is approximately a step function. Assume the renormalised band is not flat, hence $u^{(s^*)} < 2t/V$. Then Eq. (4.42) becomes

$$\lim_{\beta \to \infty} u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\epsilon_{\mathbf{k}} \leq \tilde{\mu}} (\cos k_x + \cos k_y)$$

Let FS(δ) be the Fermi surface for a given doping level δ. We denote with Int(FS) be its internal part, the Fermi sea:

$$\mathrm{FS} = \{\mathbf{k} \in \mathrm{BZ} \mid \tilde{\epsilon}_{\mathbf{k}} = \tilde{\mu}\}$$
$$\mathrm{Int(FS)} = \{\mathbf{k} \in \mathrm{BZ} \mid \tilde{\epsilon}_{\mathbf{k}} \leq \tilde{\mu}\}$$

At half filling, the Fermi surface coincides with the boundary of the MBZ.

$$\delta = 0 \quad \Longrightarrow \quad \mathrm{FS} = \partial\mathrm{MBZ}$$

This is implied by the fact that renormalised bands are antisymmetric by $\boldsymbol{\pi}$ shifts,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = -\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}\sigma}$$

The portion of an s^*-wave band outside the MBZ is mirrored with respect to the portion inside the MBZ. This implies that half the states lie in the MBZ, half out.

At $\delta > 0$ the FS expands in the region BZ \ MBZ where the form factor $\cos k_x + \cos k_y$ becomes negative. Then Eq. (4.42) becomes

$$\lim_{\beta\to\infty} u^{(s^*)}(\beta,\tilde{\mu}) \simeq \frac{1}{L_x L_y}\sum_{\epsilon_{\mathbf{k}}\le 0}|\cos k_x + \cos k_y| - \frac{1}{L_x L_y}\sum_{0<\epsilon_{\mathbf{k}}\le\tilde{\mu}}|\cos k_x + \cos k_y|$$

which, being the sum of a positive and a negative term, is maximised whenever $\tilde{\mu} = 0$, so at half filling.

As long as the band is not completely flattened by renormalisation, increasing the doping, the FS expands. Hence

$$\delta_1 > \delta_2 \quad\Longrightarrow\quad \mathrm{Int}(\mathrm{FS}_1) \subset \mathrm{Int}(\mathrm{FS}_2)$$

where FS_i is the FS associated to the doping level δ_i. Using this property we conclude

$$\tilde{t} > 0 \quad \text{and} \quad \delta_1 > \delta_2 \quad\Longrightarrow\quad \lim_{\beta\to\infty} u^{(s^*)}(\beta,\tilde{\mu})\Big|_{\delta_1} < \lim_{\beta\to\infty} u^{(s^*)}(\beta,\tilde{\mu})\Big|_{\delta_2}$$

This proves that $u^{(s^*)}$ decreases with doping at low temperature.

This is the first expected behaviour of the self-consistent functional $u^{(s^*)}$: if there exists a region of parametric space where the shrinking effect is not strong enough to flatten the band, a rise in the doping level must lower $u^{(s^*)}$. Such a region must exist: at least all the segment $(V = 0, \delta)$ is included, since on said region of parametric space the HF hamiltonian reduces to the that of the free Fermi gas.

We conclude this section with the analytic computation of $u^{(s^*)}$ at half filling and unitary filling, both at low temperature:

$$\lim_{\beta\to\infty} u^{(s^*)}[\beta,0] = \left(\frac{2}{\pi}\right)^2 \tag{4.43}$$

$$\lim_{\beta\to\infty} u^{(s^*)}\left[\beta, 2\tilde{t}^{(s^*)}\right] = 0 \tag{4.44}$$

At unitary filling we set the chemical potential equal to the maximum energy reached by a band of amplitude $4\tilde{t}^{(s^*)}$.

Proof 4.8 (Proof of Eqns. (4.43), (4.44)).

The two proofs are:

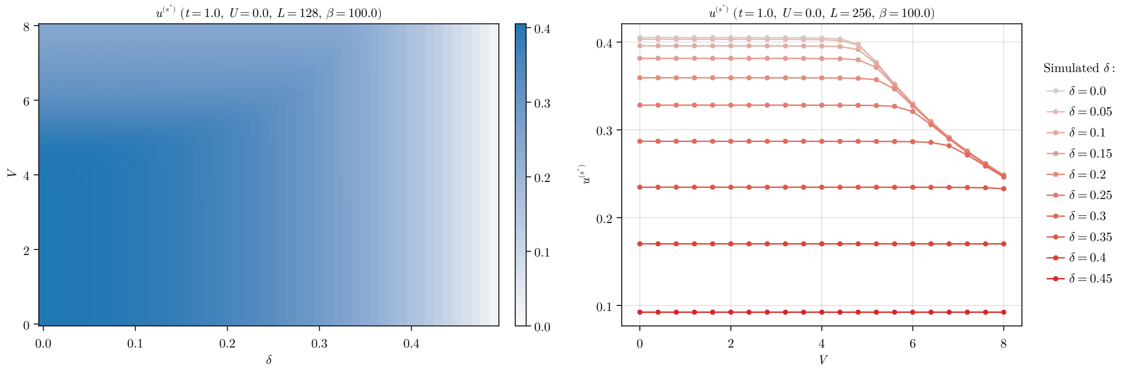
1. At half-filling, Eq. (4.42) reduces to

$$\lim_{\beta\to\infty} u^{(s^*)}(\beta,0) \simeq \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}(\cos k_x + \cos k_y)$$

In thermodynamic limit, performing integration on the MBZ shown in Fig. 4.3:

$$\begin{aligned}
\lim_{\beta\to\infty} u^{(s^*)}(\beta,0) &\simeq \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\mathrm{d}k_y\int_{-\pi+|k_y|}^{\pi-|k_y|}\mathrm{d}k_x\cos k_x + \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\mathrm{d}k_x\int_{-\pi+|k_x|}^{\pi-|k_x|}\mathrm{d}k_y\cos k_y\\
&= \frac{1}{2\pi^2}\int_{-\pi}^{0}\mathrm{d}k_y\int_{-\pi-k_y}^{\pi+k_y}\mathrm{d}k_x\cos k_x + \frac{1}{2\pi^2}\int_{0}^{\pi}\mathrm{d}k_y\int_{-\pi+k_y}^{\pi-k_y}\mathrm{d}k_x\cos k_x\\
&= \frac{1}{\pi^2}\int_{-\pi}^{0}\mathrm{d}k_y\sin(\pi+k_y) + \frac{1}{\pi^2}\int_{0}^{\pi}\mathrm{d}k_y\sin(\pi-k_y)\\
&= \left(\frac{2}{\pi}\right)^2 \approx 0.405
\end{aligned}$$

which proves Eq. (4.43).



a | Heatmap of $u^{(s^*)}$ in δV plane.

b | V-dependency of $u^{(s^*)}$ varying δ.

Figure 4.4 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase. To extract the data on the right side coarse-graining precision of the left-hand side data has been exchanged with higher lattice dimensions. As is evident, based on doping level $u^{(s^*)}$ exhibits non-trivial regions of V-independence.

2. At unitary filling we have

$$\lim_{\beta\to\infty} u^{(s^*)}\left(\beta, 2\tilde{t}^{(s^*)}\right) = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) = 0$$

which vanishes due to the structure factor periodicity by $\boldsymbol{\pi}$. This proves Eq. (4.44).

4.5 Discussion of numeric results

In this section we present and discuss the numeric results, the physical relevance of the found phenomena and the boundaries of the algorithm. For the normal phase, the HFPs "vector" defined in Sec. 2.3.2 has just one component:

$$\mathbf{v} = \left[u^{(s^*)}\right]$$

We have two RMPs:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)}$$

$$\tilde{\mu} \equiv \mu + n(2zV - U)$$

the second one being only dependent on the model parameters and not on HFPs. Finally, the phase free energy is given by Eq. (4.39). The normal phase is completely independent of U. Because of this we present the results varying the other parameters, keeping everywhere $U = 0$. We ran simulations in the (V, δ) plane, with the setup:

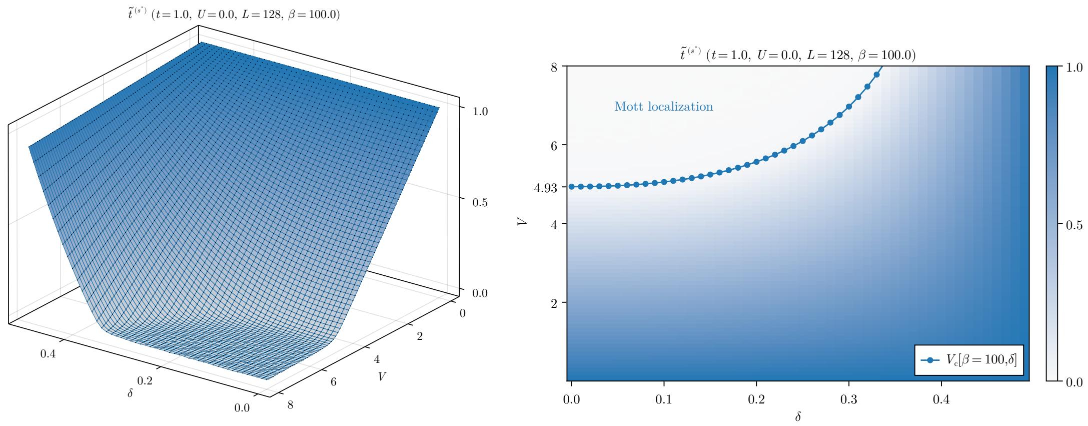
$$U = 0 \qquad \Delta n = 10^{-2} \qquad \Delta\mathbf{v} = \left[u^{(s^*)}: 5\times 10^{-4}\right]$$

with $\Delta\mathbf{v}$ the tolerance vector defined as in Sec. 2.3.2.

In Fig. 4.4a are reported the results obtained for the HFP $u^{(s^*)}$ on a 128×128 lattice and "low" temperature $\beta = 100$, varying the non-local attraction and the doping level

$$V = 0,\ 0.1,\ 0.2,\ \ldots,\ 4.0 \qquad \delta = 0,\ 0.01,\ 0.02,\ \ldots,\ 0.49$$

The expectations of Sec. 4.4 are satisfied: at low V (indicatively $V \lesssim 5$) the trend is monotonically descendent at increasing doping. As V gets bigger, however, $u^{(s^*)}$ starts decreasing in the top-left corner of Fig. 4.4a. This effect is more visible in Fig. 4.4b, where we doubled L and ran simulations with a smaller δ resolution: the convergence value of $u^{(s^*)}$ is approximately independent of V up to some point, where this behaviour is broken. Note that, as was expected from the discussion of Sec. 4.4, the half-filled convergence value of $u^{(s^*)}$ in the V-independent region is approximately given by Eq. (4.43).



a | Simulated values of $\tilde{t}^{(s^*)}$

b | Critical Mott boundary.

Figure 4.5 | Plot of the self-consistent convergence values for the HFP $u^{(s^*)}$ in the normal phase and the Mott-boundary V_c, computed as is described in text. In the right panel, the tick $V = 4.93$ is specified explicitly as a graphic confirm of the computation of Eq. (4.45). For graphic reasons the points at $V_c > 8$ have been omitted.

From Sec. 4.4 we know that $u^{(s^*)}$ decreases as we increase doping, as long as $\tilde{t}^{(s^*)} > 0$. When V gets large enough, $u^{(s^*)}$ starts decreasing because of its constraints. The net effect is the nullification of $\tilde{t}^{(s^*)}$, which is, a total localisation effect.

4.5.1 Mott localisation effect

The most interesting result we obtained for the normal phase is the possibility of localising electrons by tuning the V/δ ratio. In Fig. 4.4b, at low V the convergence value of $u^{(s^*)}$ is approximately V-independent at sufficiently low temperatures. Let $V_c[\beta, \delta]$ be the critical value such that

$$0 \leq V < V_c[\beta, \delta] \quad : \quad u^{(s^*)} \text{ is } V \text{ independent.}$$

Then within the entire segment $[0, V_c]$ the convergence value is (approximately) the same obtained for $V = 0$. We can define in first instance the critical V as the value such that the hopping correction is maximized,

$$V_c[\beta, \delta] \equiv \frac{2t}{u^{(s^*)}[\beta, \tilde{\mu}(\delta)]\big|_{V=0}}$$

When V becomes larger than this critical value, the resulting RMP $\tilde{t}^{(s^*)}$ drops to 0 and due to the constraint of Eq. (A.7) the convergence value of $u^{(s^*)}$ is forced to decrease. Note that the constraint of Eq. (A.7) does not prevent $u^{(s^*)}$ from saturating to its upper boundary from below. In Fig. 4.5a the converged values of $\tilde{t}^{(s^*)}$, extracted from the same data of Fig. 4.4a, are plotted in parametric space.

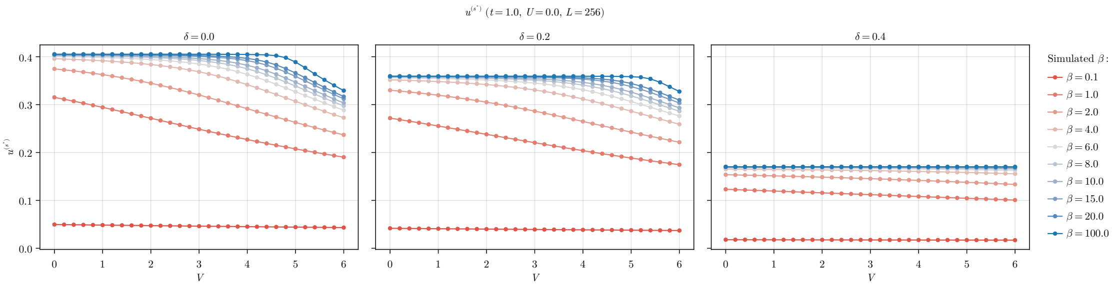
To extract the value of $V_c[\beta, \delta]$ is a non-trivial analytic problem due to the insurgence of elliptic integrals and the smearing effects of temperature. Instead, for the high-β half-filled case we can use Eq. (4.43) and get:

$$V_c[\beta \to \infty, 0] = \frac{2t}{(2/\pi)^2} \approx 4.93t \tag{4.45}$$

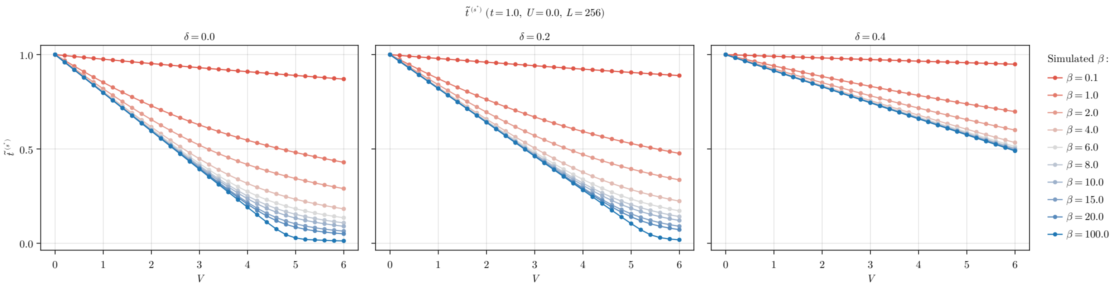
At half-filling, the effective bands amplitude reduction is so strong that small values of V (compared to t) are sufficient to nullify kinetics. In Fig. 4.5b we report, superimposed, the convergence values of the RMP $\tilde{t}^{(s^*)}$ and $V_c[\beta = 100, \delta]$ computed with the simple procedure hereby described. Our prediction of the critical value for V approximates reasonably the localisation region.

Increasing temperature Mott localisation is suppressed. Also, out precedent argument were valid only at low temperature. In Fig. 4.6a we report results for the HFP $u^{(s^*)}$ computed at decreasingly lower temperature and three example doping levels,

$$1.0 \leq \beta \leq 100.0 \qquad \delta = 0.0, 0.2, 0.4$$



a | Temperature dependence of $u^{(s^*)}$ at various dopings.



b | Temperature dependence of $\tilde{t}^{(s^*)}$ at various dopings.

Figure 4.6 | Comparative plots of the HFP $u^{(s^*)}$ (top panel) and RMP $\tilde{t}^{(s^*)}$ (bottom panel) in null-doped (left plots), "medium-doped" ($\delta = 0.2$, central plots) and "heavily-doped" ($\delta = 0.4$, right plots) contexts. Increasing temperature progressively destroys Mott localisation, reducing the bands shrinking effect.

At high temperature the V-independence region that was visible in Fig. 4.4b completely disappears, making $u^{(s^*)}$ a monotonically descendent function. In Fig. 4.6a the same dataset is plotted with focus on the RMP $\tilde{t}^{(s^*)}$. Temperature increase disrupts localisation effects, at least at small values of V. However, the bands shrinking effect is still relevant at $\beta = 1.0$, the hopping parameter approximately halving at $V \approx 5 \div 6$.

4.5.2 Solution free energy and entropy densities

In order to compare phases, we need an estimation of free energy of each. The plots of Fig. 4.7 report our findings on the normal phase free energy, computed based on Eq. (4.39). The free energy is intended in unities of t. For a non-flat band, doping increases energy because the band fills up and electrons populate higher energy levels.

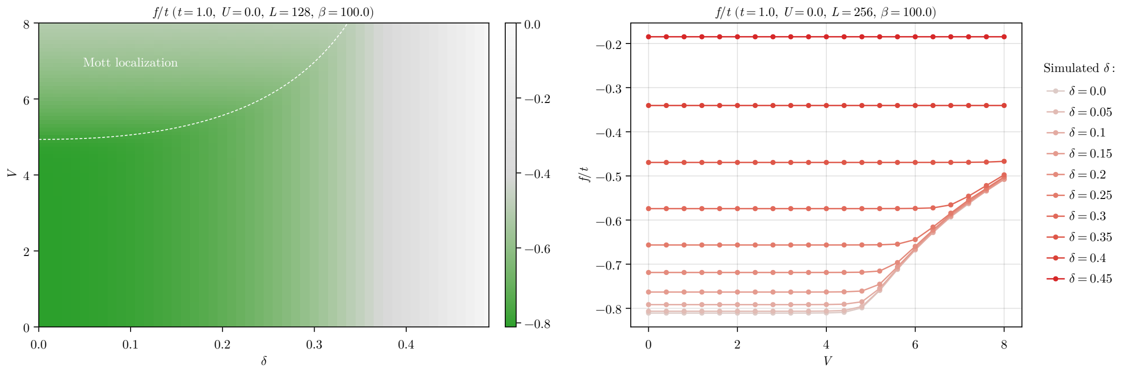
Looking the data of Fig. 4.7a, at fixed δ free energy density remains pretty much constant up to the Mott boundary. This is simply because as long as a band with non-zero amplitude is present, the HFP correction of Eq. (4.39) compensates the free energy increase due to bands shrinking,

$$
\begin{aligned}
-V\left|u^{(s^*)}\right|^2 &= -\frac{Vu^{(s^*)*}}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \\
&= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left(\epsilon_{\mathbf{k}} - \tilde{\epsilon}_{\mathbf{k}}\right) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})
\end{aligned}
$$

having we used in the second passage

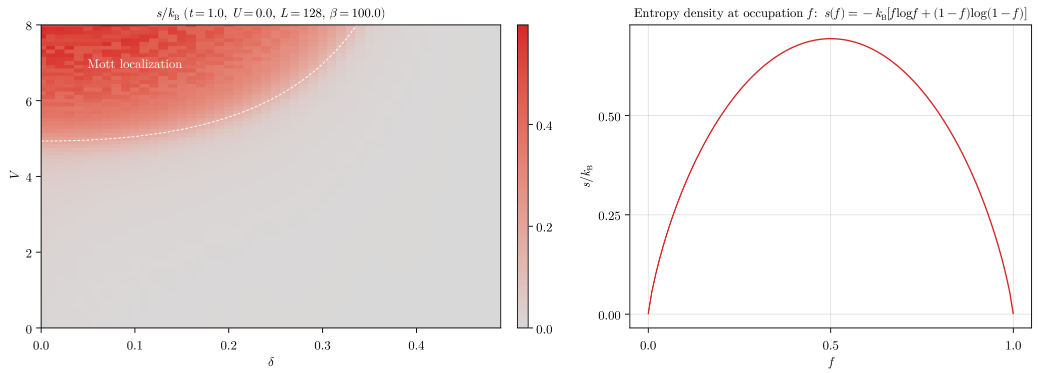
$$
\tilde{\epsilon}_{\mathbf{k}} = \tilde{\epsilon}_{\mathbf{k}} + u^{(s^*)*}V(\cos k_x + \cos k_y)
$$

Recalling the first point of the derivation of Sec. **??**, to add the HFP correction to free energy is the same as removing the *tilde* sign from single-state energies, thus using the free model ($V = 0$) free energy density expression.



a | Heatmap of f in δV plane. b | V-dependency of f varying δ.

Figure 4.7 | Free energy density of the normal phase. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 4.5b, indicating the Mott transition.



a | Heatmap of s_{MFT} in δV plane. b | V-dependency of f varying δ.

Figure 4.8 | Entropy density of the normal phase (left panel) and theoretical contribution to entropy for a state with occupation f. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 4.5b, indicating the Mott transition.

Mott localisation is energetically expensive. This is visible in both panels of Fig. 4.7. The V-dependent energy increase is an entropic effect: indeed, for a flat band:

$$f(0; \beta, \tilde{\mu}(\delta)) = \frac{1}{2}$$

at any temperature. The expression for entropy density of a free Fermi gas [29] is given by

$$s(f) = -k_{\mathrm{B}} \left[f \log f + (1 - f) \log(1 - f) \right]$$

represented in Fig. 4.8b. The maximum is located at $f = 1/2$, then a flat band realises the maximally entropic condition. The Mott insulating region can be seen as a degeneration of the FS interface to the entire BZ, a process that maximises entropy.

All of this holds, as all the rest, at low temperature: by lowering β, finite-temperature effects become more and more relevant and the statements above need to be corrected.

4.5.3 Convergence problems and mixing fine-tuning

The normal phase is simple enough to describe some of the typical convergence problems arising in HF iterative schemes anticipated in Sec. 2.3.2. Bands flattening is a source of algorithmic instability: numerics can lead to values of $u^{(s^*)}$, during iterations, slightly out of bounds with respect to Eq. (A.7). Even if the numeric effect is small, the consequence can be vast. To heal this effect and extract the actual convergence values, it is necessary to optimise the mixing parameter $g \in [0, 1]$.

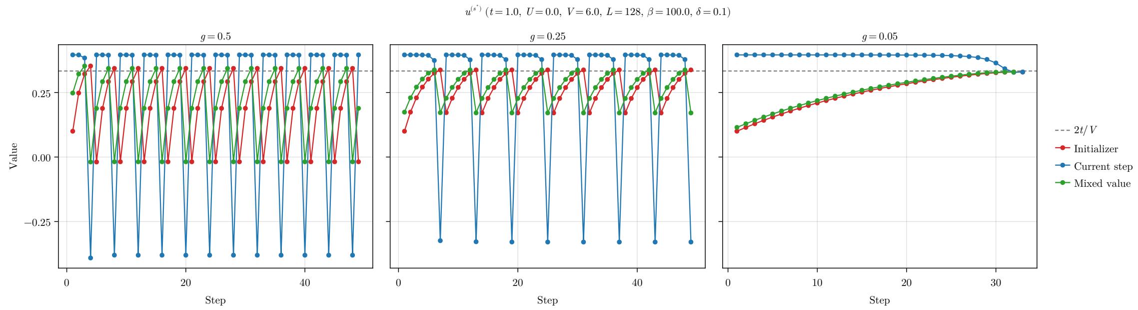


Figure 4.9 Comparison of three example tracks for the single-valued HFPs vector of the normal phase, simulated at a "Mott-deep" point.

We defined g in Sec. 2.3.2, and operatively in Eq. (2.24). This parameter controls deterministically how, from an initialiser (the HFPs vector we plug in the algorithm) and an output (the HFPs vector we get back from the algorithm) the new initialiser for the following step is generated. We show in Fig. 4.9 the evolutions at different mixing parameters of a simulation at

$$V = 6.0 \qquad \delta = 0.1 \qquad \beta = 100.0$$

This point is inside the Mott region, as can be checked in Fig. 4.4a. The HFP $u^{(s^*)}$ needs to converge at $2t/V = t/3$ in order to give complete localisation. The initialiser

$$\text{Step 1} \quad : \quad u^{(s^*)} = 0.1$$

was arbitrarily chosen. In each panel of Fig. 4.9, the red track represents the value of $u^{(s^*)}$ we plug in the self-consistency equations, the blue the result of the self-consistent equation, the green one the mixed value of the two based on different gs. The gray dashed line is the theoretical convergence target.

In the first two panels, there is an evident infinite loop occurring whenever the green line crosses the threshold $2t/V$. At those steps, the green point is out of bounds with respect to Eq. (A.7). When this happens, the following red point gives a band with inverted sign, a feature that "breaks" the self-consistency equation and generates a blue point with no physical significance.

To overcome this problem it is necessary to move "slowly". As is shown in the right panel, the choice of a small enough mixing parameter ensures convergence. In the track, each blue dot is above the target and would produce a band with inverted sign. Mixing blue and red dots with a 95% unbalance on the red ones, we ensure that the green track approaches the target from below. Note, however, that lowering g cannot be reiterated too many times. Indeed, for $g \to 0$, any point of our map becomes a fixed point.

This kind of problems can turn out to be pretty common for iHF schemes, and to understand how to knob the algorithmic parameters without sacrificing accurateness and precision, as well as avoiding computationally costly and frustrating infinite loops, was an essential part of this algorithm design.

Appendix A

Theoretical constraints on the HFPs of the normal phase

In this appendix we discuss some theoretical constraints on the two HFPs of the normal phase,

$$u^{(s^*)} \qquad u^{(d)}$$

The discussion is divided in two parts: first we consider the simpler case of unbroken C_4 symmetry, then we move to anisotropic hopping and $\mathrm{C}_4 \stackrel{\mathrm{SSB}}{\to} \mathrm{C}_2$.

Before starting, it is useful to spend a few words on the spatial structure of a mixed $s^* \otimes d$-wave band. Let $\tilde{\epsilon}_{\mathbf{k}}$ be a generic mixed band,

$$\tilde{\epsilon}_{\mathbf{k}} = \epsilon^{(s^*)}(\cos k_x + \cos k_y) + \epsilon^{(d)}(\cos k_x - \cos k_y)$$

[To be continued...]

It is the balance between the two components $\epsilon^{(s^*)}$ and $\epsilon^{(d)}$ that gives the main features of the renormalised band. Consider the s^*-wave and d-wave structure factors already presented in Fig. 3.5, and define the Magnetic Brillouin Zone (MBZ) and Nematic Brillouin Zone (NBZ) as follows:

$$\mathrm{MBZ} \equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \right\} \tag{A.1}$$

$$\mathrm{NBZ} \equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(d)} \geq 0 \right\} \tag{A.2}$$

represented in the top-left panel of Fig. A.1. In the rest of the panels of Fig. A.1 are reported examples of mixed symmetry bands, for positive components. The respective opposite sign plots are obtained by the following rules:

$$\begin{array}{llll} \epsilon^{(s^*)} < 0 & \text{and} \quad \epsilon^{(d)} < 0 & \Longrightarrow & \text{Invert colors in Fig. A.1} \\ \epsilon^{(s^*)} > 0 & \text{and} \quad \epsilon^{(d)} < 0 & \Longrightarrow & \text{Rotate plots by } \pi/2 \text{ in Fig. A.1} \\ \epsilon^{(s^*)} < 0 & \text{and} \quad \epsilon^{(d)} > 0 & \Longrightarrow & \text{Rotate plots by } \pi/2 \text{ and invert colors in Fig. A.1} \end{array}$$

Let us focus on the MBZ and the NBZ. Starting from the intersections of the two zones, both can be factored in two sectors of interest,

$$\mathrm{MBZ} = \mathcal{A} \cup \mathcal{B} \qquad \mathrm{NBZ} = \mathcal{A} \cup \mathcal{C}$$

being, accordingly to Fig. A.2,

$$\mathcal{A} \equiv \mathcal{A}_1 \cup \mathcal{A}_2 \qquad \mathcal{B} \equiv \mathcal{B}_1 \cup \mathcal{B}_2 \qquad \mathcal{C} \equiv \mathcal{C}_1 \cup \mathcal{C}_2$$

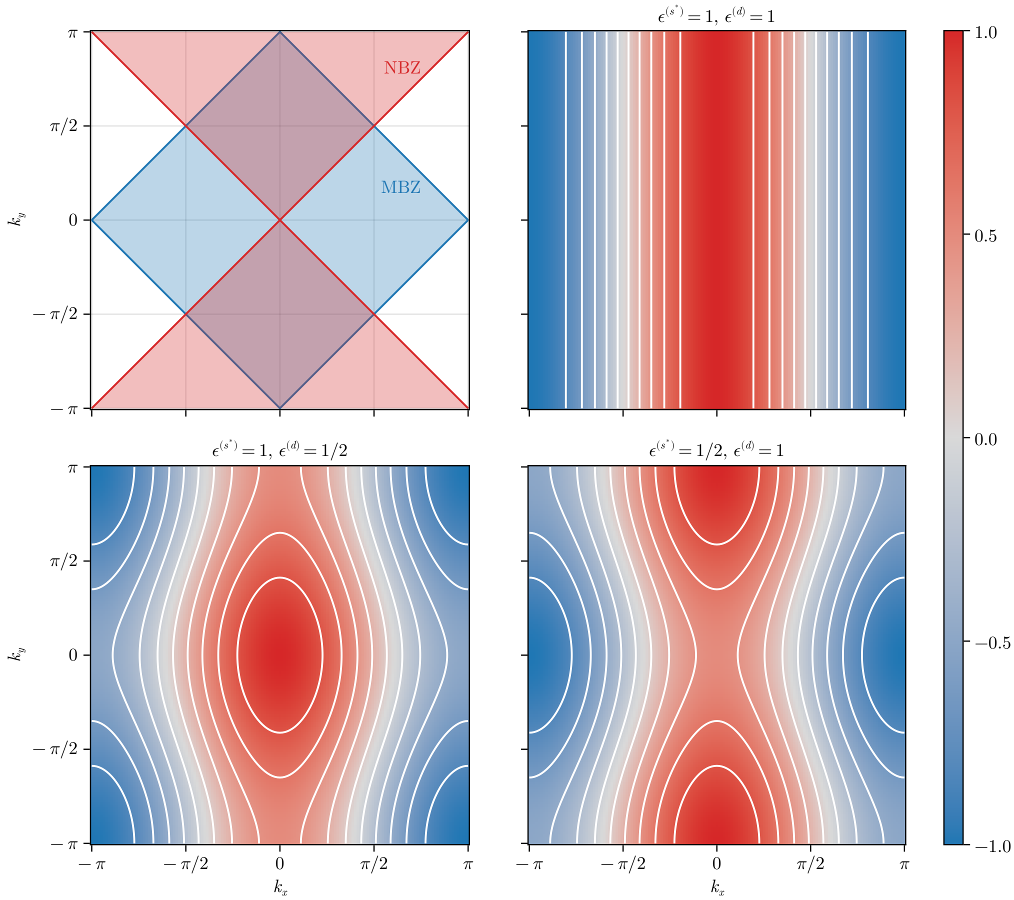


Figure A.1 Theoretical representation of mixed-symmetry nematic bands for the normal phase. The white contour represent level lines. In top-left panel, the MBZ and NBZ are depicted. While these plots only account for positive-defined components, in text is described how to map each one on its counterpart with different sign.

These domains satisfy:

$$\begin{aligned}\mathcal{A} &\equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0\right\} \\ &= \mathrm{MBZ} \cap \mathrm{NBZ}\end{aligned} \tag{A.3}$$

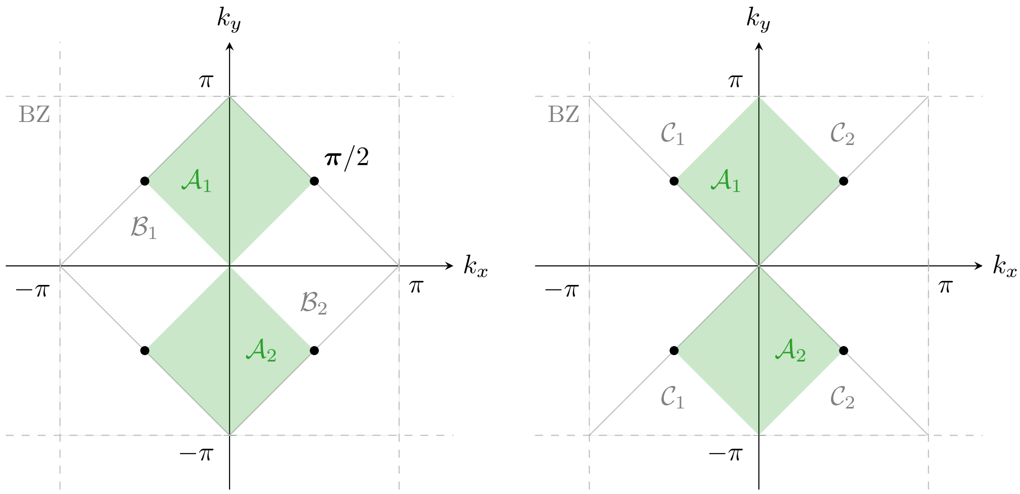
$$\begin{aligned}\mathcal{B} &\equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \leq 0\right\} \\ &= \mathrm{MBZ} \setminus \mathrm{NBZ}\end{aligned} \tag{A.4}$$

$$\begin{aligned}\mathcal{C} &\equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \leq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0\right\} \\ &= \mathrm{NBZ} \setminus \mathrm{MBZ}\end{aligned} \tag{A.5}$$

and will be used later on. Whatever the shape of the mixed band, the following equation always holds

$$|k_x| = \frac{\pi}{2} \quad \text{and} \quad |k_y| = \frac{\pi}{2} \qquad \Longrightarrow \qquad \tilde{\epsilon}_{(k_x,k_y)} = 0$$

since only cosines are involved. This identifies four node points, represented as black dots in Fig. A.2.



a | MBZ sectors. b | NBZ sectors.

Figure A.2 | Depiction of the sectors composing the MBZ (Fig. A.2a) and the NBZ (Fig. A.2b), based on the divisions shown in the top-left panel of Fig. A.1. The full green zone composes $\mathcal{A}$, the region where both the s^*-wave and d-wave structure factors are positive defined; its MBZ complementary white counterpart in the left panel gives $\mathcal{B}$, for which the s^*-wave structure factors is positive-definite while the d-wave is negative. Similarly the NBZ complementary white counterpart in the right panel gives $\mathcal{C}$, for which signs are exchanged. The black dots, at the contours intersection, are band nodes.

A.0.1 Constraints in presence of a robust C_4 symmetry and bands shrinking

Consider the simpler situation where C_4 symmetry is imposed. This implies that the renormalised bands are

$$\tilde{\epsilon}_{\mathbf{k}} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)$$

with a single self-consistency equation

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

The following chain of deductions leads us to a constraint on the possible values of $u^{(s^*)}$:

1. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \leq 0$. Let us prove this: suppose instead $u^{(s^*)} \leq 0$. For any function $g_{\mathbf{k}}$ defined on the BZ:

$$\sum_{\mathbf{k}\in\mathrm{BZ}} g_{\mathbf{k}} = \sum_{\mathbf{k}\in\mathrm{MBZ}} [g_{\mathbf{k}} + g_{\mathbf{k}+\boldsymbol{\pi}}]$$

   where we defined the vector $\boldsymbol{\pi} \equiv (\pi, \pi)$. The self-consistency equation thus reads

$$\begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}}; \beta, \tilde{\mu}) \right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \end{aligned} \tag{A.6}$$

   having we used the antisymmetry of s^*-wave functions by $\boldsymbol{\pi}$ shifts. Now, if $u^{(s^*)} \leq 0$ the renormalised band $\tilde{\epsilon}_{\mathbf{k}}$ is negative inside the MBZ and positive outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

the equality only being satisfied on the MBZ contour. Then, since inside the MBZ it holds by definition $\cos k_x + \cos k_y \geq 0$, the above equations imply

$$u^{(s^*)} \leq 0 \quad \implies \quad u^{(s^*)} \geq 0$$

To exclude the possibility of $u^{(s^*)} = 0$, it suffices to notice that from the argument above such a situation only arises when the MBZ contour degenerates to the entirety of the MBZ itself – which is, when the band is flat. But that cannot be the case for $u^{(s^*)} = 0$, as long as $t > 0$. Thus we have proven that

$$u^{(s^*)} > 0$$

2. For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \geq 2t/V$. We proceed exactly as did for the previous point, up to the passage:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$

Now, if $u^{(s^*)} \geq 2t/V$ this means the prefactor in front of the renormalised bands is positive. Then $\tilde{\epsilon}_{\mathbf{k}}$ is positive inside the MBZ and negative outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

thus we have

$$u^{(s^*)} \geq 2t/V \quad \implies \quad u^{(s^*)} \leq 0$$

which is absurd. Thus

$$u^{(s^*)} < 2t/V$$

Collecting the two points above, we have the constraint on $u^{(s^*)}$

$$\mathrm{C}_4 \quad \implies \quad 0 < u^{(s^*)} < 2t/V \tag{A.7}$$

The effective hopping is smaller than the original hopping. We can denominate this net effect "bands shrinking".

A.0.2 Natural suppression of nematicity within $\mathrm{C}_2/\mathbb{Z}_2$ symmetry

Let us extend the arguments above to a weaker symmetry, allowing for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$$

We work with the full, anisotropic band of Eq. (4.22). From the following chain of deductions we conclude that C_4 is not broken:

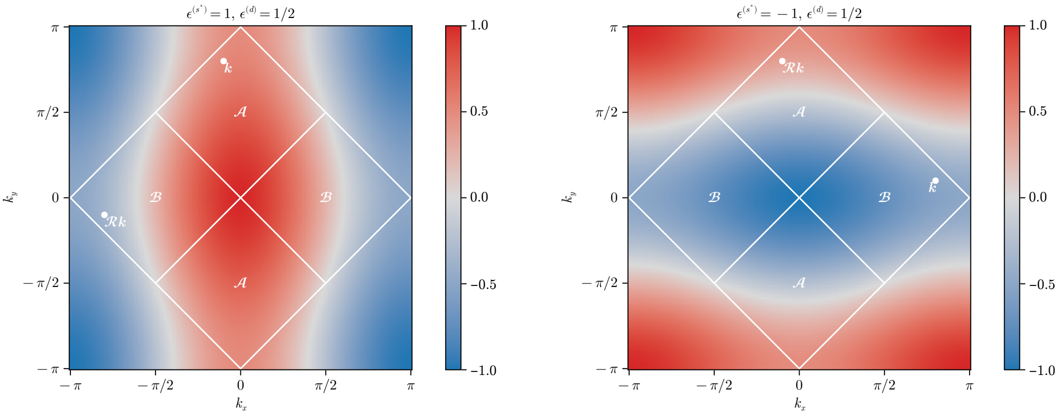
1. Let for instance

$$u^{(s^*)} \geq 2t/V > 0 \qquad u^{(d)} > 0 \quad \implies \quad \epsilon^{(s^*)} \geq 0 \qquad \epsilon^{(d)} > 0$$

In Fig. A.3a an example band is shown for $\epsilon^{(s^*)} = 1$, $\epsilon^{(d)} = 1/2$. Note that if $u^{(s^*)} = 2t/V$, the renormalised band is purely d-wave since $\epsilon^{(s^*)} = 0$.

Recall Eq. (A.6), and separate in two sums over $\mathcal{A}$ and $\mathcal{B}$:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{A}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \\ + \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{B}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$



a | Same-sign mixed bands. b | Opposite-sign mixed bands.

Figure A.3 | Theoretical example of bands with $u^{(s^*)} > 2t/V$, $u^{(d)} > 0$ (top-left) and $u^{(s^*)} < 0$, $u^{(d)} > 0$ (top right). Two example points are reported, connected by a $\pi/2$ rotation as given in Eqns. (A.10), (A.8).

Consider the sum over $\mathcal{B}$. Looking at the bottom-left panel of Fig. A.1, such a region contains nodal curves across which bands change sign. Let $\mathcal{R}$ be the clockwise $\pi/2$ rotation: then, any point in $\mathcal{B}$ can be seen as the rotation of a point in $\mathcal{A}$

$$\forall \mathbf{k}' \in \mathcal{B} \quad \exists \quad \mathbf{k} \in \mathcal{A} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \tag{A.8}$$

A graphical rendition of this idea is given in Fig. A.3a. Then we can rewrite $u^{(s^*)}$ as a single sum over $\mathcal{A}$,

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{A}} (\cos k_x + \cos k_y) \\ \times \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \tag{A.9}$$

From Eq. (A.3), $\forall \mathbf{k} \in \mathcal{A}$ the form factors $\cos k_x \pm \cos k_y$ are positive by construction. By hypothesis $\epsilon^{(s^*)} \geq 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}^{\geq 0} + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}_{\geq 0} - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0} \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \geq 0$$

thus $\tilde{\epsilon}_{\mathbf{k}} \geq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\leq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\geq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

This condition, inserted in Eq. (A.9), leads to an absurd:

$$u^{(s^*)} \geq 2t/V \qquad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

2. An identical argument, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$, can be carried out to show that

$$u^{(s^*)} \geq 2t/V > 0 \qquad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

3. The two points above imply that it is possible to have $u^{(s^*)} \geq 2t/V$ if and only if $u^{(d)} = 0$. But from Sec. A.0.1, we conclude that the upper boundary $u^{(s^*)} < 2t/V$ always holds.

4. Let for instance

$$u^{(s^*)} \leq 0 \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad \epsilon^{(s^*)} < 0 \qquad \epsilon^{(d)} > 0$$

In Fig. A.3b an example band is shown for $\epsilon^{(s^*)} = -1$, $\epsilon^{(d)} = 1/2$.

Proceed as for the first point. Similarly to Eq. (A.8), Any point in $\mathcal{A}$ can be obtained by rotating some point in $\mathcal{B}$,

$$\forall \mathbf{k}' \in \mathcal{A} \quad \exists \quad \mathbf{k} \in \mathcal{B} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \tag{A.10}$$

An equation similar to Eq. (A.9) holds, the sum being performed over $\mathcal{B}$

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathcal{B}} (\cos k_x + \cos k_y) \\ \times \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \tag{A.11}$$

From Eq. (A.4), $\forall \mathbf{k} \in \mathcal{B}$ the form factor $\cos k_x + \cos k_y$ is positive by construction; the form factor $\cos k_x - \cos k_y$ is negative by construction. By hypothesis $\epsilon^{(s^*)} < 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}^{\leq 0} + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\leq 0} \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}_{\leq 0} - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\leq 0} \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \leq 0$$

thus $\tilde{\epsilon}_{\mathbf{k}} \leq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\geq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\leq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \end{aligned}$$

This condition, inserted in Eq. (A.9), leads to an absurd:

$$u^{(s^*)} \leq 0 \qquad u^{(d)} > 0 \qquad \Longrightarrow \qquad u^{(s^*)} > 0$$

5. can be carried out to show that

$$u^{(s^*)} \leq 0 \qquad u^{(d)} < 0 \qquad \Longrightarrow \qquad u^{(s^*)} > 0$$

6. The two points above imply that it is possible to have $u^{(s^*)} \leq 0$ if and only if $u^{(d)} = 0$. But from Sec. A.0.1, we conclude that the lower boundary $u^{(s^*)} > 0$ always holds.

From the chain of derivations above, we can conclude self-consistently that also in the nematic structure the constraint of Eq. (A.7),

$$0 < u^{(s^*)} < 2t/V$$

holds. In the s^*-wave channel, partial breaking of rotational symmetry is not enough to modify the main effect identified under C_4 symmetry, “bands shrinking”. What is missing is the behaviour of the d-wave channel when said effect takes place. Continuing the enumeration above,

7. Let $0 < u^{(s^*)} < 2t/V$ and $u^{(d)} > 0$. Then $\epsilon^{(d)} > 0$. The self-consistency equation for $u^{(d)}$ reads

$$
\begin{aligned}
u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{NBZ}} (\cos k_x - \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right]
\end{aligned}
$$

where we recognised, as is shown in Fig. A.2b, that BZ $\setminus$ NBZ $= \mathcal{R}$NBZ. It holds:

$$
\forall \mathbf{k} \in \mathrm{NBZ} \quad : \quad
\begin{aligned}
\tilde{\epsilon}_{\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \\
\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0}
\end{aligned}
$$

thus $\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} < \tilde{\epsilon}_{\mathbf{k}}$, and inserting this fact in the self-consistency equation above we get

$$
0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(d)} < 0
$$

which is absurd.

8. An identical argument, with inverted inequalities, can be carried out to show that

$$
0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(d)} > 0
$$

9. The two points above imply that it is possible to satisfy the constraint $0 < u^{(s^*)} < 2t/V$ if and only if $u^{(d)} = 0$.

All the deductions above indicate that $u^{(d)} = 0$ everywhere. Nematic effects are always absent,

$$
0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \tag{A.12}
$$

This is physically expected but non trivial. The normal phase does not break C_4 and we can ignore $u^{(d)}$ at all. In next section we identify the expected self-consistent value for $u^{(s^*)}$.

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